

## Estimation of pure component properties Part 2. Estimation of critical property data by group contribution

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### Abstract

A new group contribution method for the estimation of critical property data has been developed. The method is based on structural group definitions, with minor modifications, of a method recently published for the estimation of the normal boiling point by Nannoolal et al. [Y. Nannoolal, J. Rarey, D. Ramjugernath, W. Cordes, *Fluid Phase Equilib.* 226 (2004) 45–63]. Special care was taken to ensure physically meaningful extrapolation to large and multifunctional compounds, as it could be shown that such extrapolation may lead to very improbable or unrealistic results in the case of some methods published earlier. Regression of group increments was not performed simultaneously but with great care one group at a time. With all other group increments fixed, all data for components with a specific group led to approximately the same group increment. This allowed one to determine group increments for cases where only one reliable experimental value was available. For the case of more than one strongly associating group, a group interaction contribution was required as these groups do not observe simple additivity. The estimation of the critical temperature requires knowledge of the normal boiling point. Critical pressure and volume can be estimated from chemical structure alone. The performance of the new model has been compared with 10 well-known estimation methods from literature and the results indicate that the new model is significantly more reliable. To enable comparison, chemical family definitions have been developed that allow one to automatically classify new components and thus, inform the user about the expected reliability of the different methods for a component of interest. Chemical family definitions are based on the kind and frequency of the different structural groups in the molecule. In addition to these advantages, the range of applicability of the new method is larger than that of comparable methods.

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### 1. Introduction

Critical property data are of great practical importance as they are the basis for the estimation of a large variety of thermodynamic properties using the corresponding states principle. In addition, critical temperature and pressure data provide valuable information for the regression and prediction of vapor pressures at high temperature and are required by equations of state for the description of pure component and mixture behavior.

At the critical point, all components show the same striking anomalies like infinite heat capacity and compressibility. Fluid properties in this state are completely dominated by long range fluctuations and density correlation. The critical temperature and

pressure are extremely well defined. Due to the infinite compressibility, experimental critical densities are usually of lower reliability.

Unfortunately, most components are not sufficiently stable at or near the critical temperature, and as a result experimental measurements of their critical properties are extremely difficult, if not impossible. It is therefore vital that prediction methods be developed which are capable of not only reasonably accurate predictions, but which are also reliable with a low probability of failure when applied to extrapolation. Unluckily, the critical point is difficult to capture by molecular simulation due to the very large autocorrelation length at and near this state.

Since the first developments of group contribution methods by Riedel [1] in 1949 and Lydersen [2] in 1955, a large number of methods has been developed for the estimation of critical property data. While various different approaches can be found

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in literature, the use of the group contribution concept still seems to provide the most reliable and easy to obtain results.

The objective of this work was to develop a new estimation method for the critical temperature, pressure and volume of organic compounds with a wide range of applicability and to give a detailed analysis of its performance compared to ten well-known previously published methods.

Reliability analysis of such estimation methods is of significant practical importance as a tool to assess process model reliability in chemical process design and optimization. Different authors have reported extensive tests of critical property estimation methods in the past based on a smaller set of data, fewer methods and less differentiation with respect to classes of chemically similar compounds. A very detailed comparative study of four different estimation methods was recently published by Yan et al. [3].

## 2. Overview on available methods and motivation

A variety of estimation methods for critical property data are available in the open literature. A broad overview of these methods together with a detailed discussion of their reliability was given by Poling et al. [4]. In addition, several authors evaluated the performance of models utilizing a large common set of experimental data [3]. In this paper the comparison to other generally applicable methods is restricted to those which are based on the group contribution concept and are applicable over a wide range of components. Table 1 gives an overview on the timeline of major previous developments in group contribution methods for critical properties. Due to their practical and theoretical importance, estimation of critical properties has attracted much interest of researchers from all over the world. Critical property estimation methods restricted to individual classes of components (alkanes, alkenes, etc.) were excluded from Table 1.

Table 1  
Important group contribution methods for the estimation of critical property data

| Year | Reference                                    | Additional information<br>( $T_c$ estimation) | Abbreviation |
|------|----------------------------------------------|-----------------------------------------------|--------------|
| 1949 | Riedel [1]                                   | –                                             | RI           |
| 1955 | Lydersen [2]                                 | $T_b$                                         | LD           |
| 1978 | Ambrose [5]                                  | $T_b$                                         | –            |
| 1979 | Ambrose [6]                                  | $T_b$                                         | AB           |
| 1980 | Daubert [7]                                  | $T_b$                                         | DB           |
| 1982 | Fedors [8]                                   | –                                             | –            |
| 1984 | Joback [9]                                   | $T_b$                                         | –            |
| 1984 | Klincewicz and Reid [10]                     | $T_b$                                         | KR           |
| 1986 | Somayajulu [11]                              | $T_b$                                         | SJ           |
| 1987 | Joback and Reid [12]                         | $T_b$                                         | JR           |
| 1994 | Constantinou and Gani [13]                   | –                                             | CG           |
| 1995 | Tu [14]                                      | –                                             | CT           |
| 1996 | Wilson and Jasperson [15]                    | $T_b$                                         | WJ           |
| 1999 | Marrero-Morejon and Pardillo-Fontdevila [16] | $T_b$                                         | MP           |
| 2001 | Marrero-Morejon and Gani [17]                | $T_b$                                         | –            |
| 2001 | Wen and Quiang [18]                          | $T_b$                                         | WQ           |

In addition to the methods given in Table 1, numerous publications cover the use of QSPR (quantitative structure property relation) correlations and popular mathematical methods like neural networks for critical property estimation. While the correlative power of these approaches has been demonstrated in many cases, the extrapolative ability of these methods is not convincing, especially to conditions well outside their correlative range.

Classical estimation techniques can be divided into those which require only the molecular structure and others which require further relevant properties. For typical organic compounds of interest, the ratio of  $T_c/T_b$  is often within the range of  $1.4 \pm 0.3$ . Thus, knowledge of the normal boiling temperature greatly simplifies critical temperature estimation. If experimental normal boiling point or vapor pressure information is not available, group contribution estimation for this auxiliary property can be employed. These are usually of better quality and have a greater range of applicability due to the much larger set of experimental data available for these properties e.g., reliable critical temperatures can be found for approximately only 600 components, whilst experimental normal boiling point data in the open literature cover more than 18 000 substances.

Besides the simple group additivity schemes, a number of more complex estimation routes using topological indices [5,6] or bond interactions [16] have also been employed in this comparative study. Most of the available methods have been implemented in the software package Artist [19]. Estimation results were compared with a critically evaluated database in order to develop an expert system for the selection of the best model for a specific type of component. Model implementations were carefully verified, partly in cooperation with the authors.

The new estimation method developed is based on our experience with previous work on normal boiling temperatures [20,21] and vapor pressures [22]. As our preliminary work [23] showed significant improvements with respect to the published methods, it was decided to also invest further effort into the development of a new group contribution method for critical property data. In addition to a detailed description of the new method, this paper contains extensive comparative test results for 10 literature methods (The RI and WJ methods have not been included due to historical reasons for the RI method and the fact that the WJ method requires additional information apart from structure and boiling point, viz. density and is outside the scope of this work). These should be of great value for users who rely on critical property estimation for process simulation, risk assessment or environmental models.

## 3. Development of the new method and results

The new method employs a slightly improved fragmentation scheme compared to the method for the estimation of normal boiling temperature published earlier [21]. The list of structural groups for the new method, second order corrections and interacting groups are given in Tables 2–4, respectively. All modified groups and corrections from the previous work are listed in

Table 2  
Group definitions (ID, identification number; PR, priority)

| Group                          | Description                                                                                                             | Name                                          | ID, PR  | Occurs e.g. in                                                       |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|---------|----------------------------------------------------------------------|
| Periodic Group <sup>a</sup> 17 |                                                                                                                         |                                               |         |                                                                      |
| Fluorine                       |                                                                                                                         |                                               |         |                                                                      |
| F—                             | F— connected to nonaromatic C or Si                                                                                     | F—(C,Si)                                      | 19, 87  | 2-Fluoropropane, trimethylfluorosilane                               |
|                                | F— connected to C or Si with at least one F or Cl neighbor and one other atom                                           | F—((C,Si)—([F,Cl]))—a                         | 22, 84  | 1-Chloro-1,2,2,2-tetrafluoroethane [R124], difluoromethylsilane      |
|                                | F— connected to C or Si already substituted with at least one F and two other atoms                                     | F—((C,Si)—([F]))—b                            | 21, 81  | 1,1,1-Trifluoroethane, 2,2,3,3-tetrafluoropropionic acid             |
|                                | F— connected to C or Si already substituted with at least one Cl and two other atoms                                    | F—((C,Si)—(Cl))—b                             | 102, 82 | Trichlorofluoromethane [R11], 1,1-dichloro-1-fluoroethane [R141B]    |
|                                | F— connected to C or Si already substituted with two F or Cl                                                            | F—((C,Si)—([F,Cl] <sub>2</sub> ))             | 23, 83  | 1,1,1-Trifluorotoluene, 2,2,2-trifluoroethanol, trifluoroacetic acid |
|                                | F— connected to an aromatic C                                                                                           | F—(C(a))                                      | 24, 86  | Fluorobenzene, 4-fluoroaniline                                       |
|                                | F— on a C=C (vinylfluoride)                                                                                             | F—C=C<                                        | 20, 85  | Vinyl fluoride, trifluoroethene, perfluoropropylene                  |
| Chlorine                       |                                                                                                                         |                                               |         |                                                                      |
| Cl—                            | Cl— connected to C or Si not already substituted with F or Cl                                                           | Cl— (C,Si)                                    | 25, 71  | Butyl chloride, 2-chloroethanol, chloroacetic acid                   |
|                                | Cl— connected to C or Si already substituted with one F or Cl                                                           | Cl—((C,Si)—([F,Cl]))                          | 26, 70  | Dichloromethane, dichloroacetic acid, dichlorosilane                 |
|                                | Cl— connected to C or Si already substituted with at least two F or Cl                                                  | Cl—((C,Si)—([F,Cl] <sub>2</sub> ))            | 27, 68  | Ethyl trichloroacetate, trichloroacetoneitrile                       |
|                                | Cl— connected to an aromatic C                                                                                          | Cl—(C(a))                                     | 28, 72  | Chlorobenzene                                                        |
|                                | Cl— on a C=C (vinylchloride)                                                                                            | Cl—C=C<                                       | 29, 69  | Vinyl chloride                                                       |
|                                | COCl— connected to C (acid chloride)                                                                                    | COCl—                                         | 77, 18  | Acetyl chloride, phenylacetic acid chloride                          |
| Bromine                        |                                                                                                                         |                                               |         |                                                                      |
| Br—                            | Br— connected to a non-aromatic C or Si                                                                                 | Br—(C,Si(na))                                 | 30, 65  | Ethyl bromide, bromoacetone                                          |
|                                | Br— connected to an aromatic C                                                                                          | Br—(C(a))                                     | 31, 66  | Bromobenzene                                                         |
| Iodine                         |                                                                                                                         |                                               |         |                                                                      |
| I—                             | I— connected to C or Si                                                                                                 | I—(C,Si)                                      | 32, 63  | Ethyl iodide, 2-iodotoluene                                          |
| Periodic Group 16              |                                                                                                                         |                                               |         |                                                                      |
| Oxygen                         |                                                                                                                         |                                               |         |                                                                      |
| —OH                            | —OH for aliphatic chains with less than five C (cannot be connected to aromatic groups)                                 | —OH (<C <sub>5</sub> ) (z)                    | 36, 92  | Ethanol, propanediol                                                 |
|                                | —OH connected to C or Si substituted with one C or Si in an at least five C or Si containing chain (primary alkanols)   | —OH (>C <sub>4</sub> ) (z)                    | 35, 88  | 1-Nonanol, tetrahydrofurfuryl alcohol, ethylene cyanohydrin          |
|                                | —OH connected to a C or Si substituted with two C or Si in at least three C or Si containing chain (secondary alkanols) | HO—((C,Si) <sub>2</sub> H—(C,Si)—(C,Si)—) (z) | 34, 90  | 2-Butanol, cycloheptanol                                             |
|                                | —OH connected to C which has four non-hydrogen neighbors (tertiary alkanols)                                            | —OH <i>tert</i>                               | 33, 91  | <i>tert</i> -Butanol, diacetone alcohol                              |
|                                | —OH connected to an aromatic C (phenols)                                                                                | HO—(Ca) (z)                                   | 37, 89  | Phenol, methyl salicylate                                            |
| —O—                            | —O— connected to two neighbors which are each either C or Si (ethers)                                                   | (C,Si)—O—(C,Si) (z)                           | 38, 94  | Diethyl ether, 1,4-dioxane                                           |

Table 2 (Continued)

| Group                 | Description                                                                     | Name                         | ID, PR  | Occurs e.g. in                                     |
|-----------------------|---------------------------------------------------------------------------------|------------------------------|---------|----------------------------------------------------|
|                       | —O— in an aromatic ring with aromatic C as neighbors                            | (C(a))—O(a)—(C(a)) (z)       | 65, 93  | Furan, furfural                                    |
| —CHO                  | CHO— connected to non-aromatic C (aldehydes)                                    | CHO—(Cna) (z)                | 52, 53  | Acetaldehyde, pentanedial                          |
|                       | CHO— connected to aromatic C (aldehydes)                                        | CHO—(Ca) (z)                 | 90, 52  | Furfural, benzaldehyde                             |
| >C=O                  | —CO— connected to two non-aromatic C (ketones)                                  | O=C<(Cna) <sub>2</sub> (z)   | 51, 4   | Acetone, methyl cyclopropyl ketone                 |
|                       | —CO— connected to two C with at least one aromatic C (ketones)                  | (O=C<(C) <sub>2</sub> )a (z) | 92, 54  | Acetophenone, benzophenone                         |
|                       | —CO connected to N                                                              | >N—(C=O)                     | 109, 39 | Methyl thioacetate                                 |
|                       | —CO connected to two N (urea)                                                   | >N—(C=O)—N<                  | 100, 3  | Urea-1,1,3,3-tetramethyl                           |
|                       | 1,2-diketone (Do not fragment)                                                  | O=C—C=O                      | 118, 1  | 2,3-Butandione                                     |
| O=C(—O—) <sub>2</sub> | Non-cyclic carbonate diester                                                    | O=C(—O—) <sub>2</sub>        | 79, 14  | Dimethyl carbonate                                 |
| COOH—                 | —COOH connected to C (carboxylic acid)                                          | COOH—(C) (z)                 | 44, 23  | Acetic acid                                        |
| —COO—                 | HCOO— connected to C (formic acid ester)                                        | HCOO—(C) (z)                 | 46, 26  | Ethyl formate, phenyl formate                      |
|                       | —COO— connected to two C (ester) in a chain                                     | (C)—COO—(C) (z)              | 45, 24  | Ethyl acetate, vinyl acetate                       |
|                       | —COO— in a ring, C is connected to C (lactone)                                  | —C(r)OO— (z)                 | 47, 25  | $\epsilon$ -Caprolactone, crotonolactone           |
| —OCOO—                | —CO connected to two O (carbonates)                                             | —OCOO—                       | 103, 33 | Propylene carbonate, 1,3 dioxolan-2-one            |
| —OCON<                | —CO connected to O and N (carbamate)                                            | —OCON<                       | 99, 2   | Trimethylsilyl methylcarbamate                     |
| >(OC <sub>2</sub> )<  | >(OC <sub>2</sub> )< (epoxide)                                                  | >(OC <sub>2</sub> )< (z)     | 39, 50  | Propylene oxide                                    |
| —CO—O—CO—             | Anhydride connected to two C                                                    | —C=O—O—C=O—                  | 76, 11  | Acetic anhydride, butyric anhydride                |
|                       | Cyclic anhydride connected to two C connected by a double bond or aromatic bond | (—C=O—O—C=O—)r               | 96, 10  | Maleic anhydride, phthalic anhydride               |
| —O—O—                 | Peroxide                                                                        | —O—O—                        | 94, 31  | Di- <i>tert</i> -butylperoxide                     |
| Sulphur               |                                                                                 |                              |         |                                                    |
| —S—S—                 | —S—S— (disulfide) connected to two C                                            | (C)—S—S—(C)                  | 55, 51  | Dimethyldisulfide, 1,2-dicyclopentyl-1,2-disulfide |
| —SH                   | —SH connected to C (thiols, mercaptanes)                                        | SH—(C) (z)                   | 53, 73  | 1-Propanethiol                                     |
| —S—                   | —S— connected to two C (thioether)                                              | (C)—S—(C) (z)                | 54, 74  | Methyl ethyl sulfide                               |
|                       | —S— in an aromatic ring (aromatic thioether)                                    | —S(a)— (z)                   | 56, 75  | Thiazole, thiophene                                |
| —SO <sub>2</sub> —    | Non-cyclic sulfone connected to two C (sulfone)                                 | (C)—SO <sub>2</sub> —(C)     | 82, 17  | Sulfolane, divinylsulfone                          |
| >SO <sub>4</sub>      | Sulfates                                                                        | >SO <sub>4</sub>             | 104, 34 | Dimethyl sulfate                                   |
| —SO <sub>2</sub> N<   | —S(=O) <sub>2</sub> connected to N (sulfonamide)                                | —SO <sub>2</sub> N<          | 105, 35 | <i>N,N</i> -diethylmethanesulfonamide              |
| >S=O                  | Sulfoxide                                                                       | >S=O                         | 107, 37 | 1,4-Thioxane-S-oxide, tetramethylene sulfoxide     |
| SCN—                  | SCN— (isothiocyanate) connected to C                                            | SCN—(C)                      | 81, 19  | Allyl isothiocyanate                               |
| Selenium              |                                                                                 |                              |         |                                                    |
| >Se<                  | >Se< connected to at least one C or Si                                          | >Se<                         | 116, 46 | Dimethyl selenide                                  |
| Periodic Group 15     |                                                                                 |                              |         |                                                    |
| Nitrogen              |                                                                                 |                              |         |                                                    |
| NH <sub>2</sub> —     | NH <sub>2</sub> — connected to either C or Si (primary amine)                   | NH <sub>2</sub> —(C,Si) (z)  | 40, 96  | Hexylamine, ethylenediamine                        |
|                       | NH <sub>2</sub> — connected to an aromatic C (aromatic primary amine)           | NH <sub>2</sub> —(Ca) (z)    | 41, 95  | Aniline, benzidine                                 |
| —NH—                  | —NH— connected to two C or Si neighbors (secondary amine)                       | (C,Si)—NH—(C,Si) (z)         | 42, 100 | Diethylamine, diallyl amine                        |

|                     |                                                                            |                                            |         |                                                         |
|---------------------|----------------------------------------------------------------------------|--------------------------------------------|---------|---------------------------------------------------------|
|                     | —NH— connected to two C or Si neighbors in a ring (cyclic secondary amine) | (C,Si)r—NH—(Ca,Si)r (z)                    | 97, 99  | Morpholine, pyrrolidine                                 |
| >N—                 | >N— connected to three C or Si neighbors (tertiary amine)                  | (C,Si) <sub>2</sub> >N—(C,Si)              | 43, 101 | N,N-dimethylaniline, nicotine                           |
|                     | >N< connected to four C or Si (quaternary amine)                           | (C,Si) <sub>2</sub> >N<(C,Si) <sub>2</sub> | 101, 32 | N,N,N,N-tetramethylmethylenediamine                     |
| =N—                 | Double bonded amine connected to at least a C or Si                        | (C,Si)=N—                                  | 91, 102 | Acetonin                                                |
| —N—                 | Aromatic —N— in a five membered ring, free electron pair                   | =N(a)— (r5) (z)                            | 66, 98  | Piperidine, thiazole                                    |
| =N—                 | Aromatic =N— in a six membered ring                                        | =N(a)— (r6) (z)                            | 67, 97  | Pyridine, nicotine                                      |
| C≡N—                | —C≡N (cyano-group) connected to C (cyanide)                                | (C)—C≡N (z)                                | 57, 55  | Acetonitrile, 2,2'-dicyano diethyl sulfide              |
|                     | —C≡N (cyano-group) connected to N (cyanamide)                              | (N)—C≡N                                    | 111, 41 | Dimethylcyanamide                                       |
|                     | —C≡N (cyano-group) connected to S (thiocyanate)                            | (S)—C≡N                                    | 108, 38 | Methyl thiocyanate                                      |
| CNCNC—r             | Imadizole (aromatic 5-ring)                                                | ..=CNC=NC=..                               | 106, 36 | 1-Methyl-1-imadizole                                    |
| —CONH<              | —CONH <sub>2</sub> (amide)                                                 | —CONH <sub>2</sub>                         | 50, 27  | Acetamide                                               |
|                     | —CONH— (monosubstituted amide)                                             | —CONH—                                     | 49, 48  | N-Methylformamide, 6-caprolactam                        |
|                     | —CON< (disubstituted amide)                                                | —CON<                                      | 48, 49  | N,N-Dimethylformamide (DMF)                             |
| OCN—                | OCN— connected to C or Si (isocyanate)                                     | OCN— (z)                                   | 80, 28  | Butylisocyanate, hexamethylene diisocyanate             |
| ONC—                | ONC— (oxime)                                                               | ONC—                                       | 75, 29  | Methyl ethyl ketoxime                                   |
| —ON=                | —ON=connected to C or Si (isoazole)                                        | —ON=(C,Si)                                 | 115, 45 | Isoxazole, 5-phenyl isoxazole                           |
| NO <sub>2</sub> —   | Nitrites (esters of nitrous acid)                                          | O=N—O—(C)                                  | 74, 22  | Ethyl nitrite, nitrous acid methyl ester                |
|                     | NO <sub>2</sub> — connected to aliphatic C                                 | NO <sub>2</sub> —(C)                       | 68, 20  | 1-Nitropropane                                          |
|                     | NO <sub>2</sub> — connected to aromatic C                                  | NO <sub>2</sub> —(C(a)) (z)                | 69, 21  | Nitrobenzene                                            |
| NO <sub>3</sub> —   | Nitrate (esters of nitric acid)                                            | NO <sub>3</sub> —                          | 72, 13  | N-butylnitrate, 1,2-propanediol dinitrate               |
| Phosphorous         |                                                                            |                                            |         |                                                         |
| >P(O—) <sub>3</sub> | Phosphate triester                                                         | PO(O—) <sub>3</sub>                        | 73, 9   | Triethyl phosphate, tris-(2,4-dimethylphenyl) phosphate |
| >P<                 | Phosphorus connected to at least one C or S (phosphine, phosphane)         | >P—(C,Si)                                  | 113, 43 | Triphenylphosphine, triethylphosphane                   |
| Arsine              |                                                                            |                                            |         |                                                         |
| AsCl <sub>2</sub> — | AsCl <sub>2</sub> connected to C                                           | AsCl <sub>2</sub> —                        | 84, 16  | Ethylarsenic dichloride                                 |
| Periodic Group 14   |                                                                            |                                            |         |                                                         |
| Carbon              |                                                                            |                                            |         |                                                         |
| —CH <sub>3</sub>    | CH <sub>3</sub> — not connected to either N, O, F or Cl                    | CH <sub>3</sub> —(ne)                      | 1, 105  | Decane                                                  |
|                     | CH <sub>3</sub> — connected to either N, O, F or Cl                        | CH <sub>3</sub> —(e)                       | 2, 103  | Dimethoxymethane, methyl butyl ether                    |
|                     | CH <sub>3</sub> — connected to an aromatic atom (not necessarily C)        | CH <sub>3</sub> —(a)                       | 3, 104  | Toluene, <i>p</i> -methyl-styrene                       |
| —CH <sub>2</sub> —  | —CH <sub>2</sub> — in a chain                                              | —C(c)H <sub>2</sub> —                      | 4, 112  | Butane                                                  |
|                     | —CH <sub>2</sub> — in a ring                                               | —C(r)H <sub>2</sub> —                      | 9, 113  | Cyclopentane                                            |
| >CH—                | >CH— in a chain                                                            | >C(c)H—                                    | 5, 119  | 2-Methylpentane                                         |
|                     | >CH— in a ring                                                             | >C(r)H—                                    | 10, 118 | Methylcyclohexane                                       |
| >C<                 | >C< in a chain                                                             | >C(c)<                                     | 6, 121  | Neopentane                                              |
|                     | >C< in a chain connected to at least one aromatic carbon                   | >C(c)< (a)                                 | 8, 109  | Ethylbenzene, diphenylmethane                           |
|                     | >C< in a chain connected to at least one F, Cl, N or O                     | >C(c)< (e)                                 | 7, 108  | Ethanol                                                 |
|                     | >C< in a ring                                                              | >C(r)<                                     | 11, 120 | Beta-pinene                                             |
|                     | >C< in a ring connected to at least one aromatic carbon                    | >C(r)< (Ca)                                | 14, 107 | Indene, 2-methyl tetralin                               |

Table 2 (Continued)

| Group                   | Description                                                                                                        | Name                        | ID, PR   | Occurs e.g. in                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------|----------|------------------------------------------------|
| $=C(a)<$                | $>C<$ in a ring connected to, at least one N or O which are not part of the ring, or one Cl or F                   | $>C(r)<$ (e,c)              | 12, 110  | Cyclopentanol, menthol                         |
|                         | $>C<$ in a ring connected to at least one N or O which are part of the ring                                        | $>C(r)<$ (e,r)              | 13, 111  | Morpholine, nicotine                           |
|                         | Aromatic $=CH-$                                                                                                    | $=C(a)H-$                   | 15, 106  | Benzene                                        |
|                         | Aromatic $=C<$ not connected to either O,N,Cl or F                                                                 | $=C(a)<$ (ne)               | 16, 117  | Ethylbenzene, benzaldehyde                     |
|                         | Aromatic $=C<$ with three aromatic neighbors and three aromatic bonds                                              | $(a)=C(a)<_2(a)$            | 18, 115  | Naphthalene, quinoline                         |
|                         | Aromatic $=C<$ connected to either O,N,Cl or F                                                                     | $=C(a)<$ (e)                | 17, 114  | Aniline, phenol                                |
| $>C=C<$                 | Aromatic $=C<$ with three aromatic neighbors and two aromatic bonds (aliphatic bridge bond between aromatic rings) | $C(a)=C(a)<C_2(a)$ (bridge) | 214, 116 | Biphenyl, <i>m</i> -terphenyl                  |
|                         | $H_2C=C<$ (1-ene)                                                                                                  | $H_2C(c)=C<$                | 61, 57   | 1-Hexene                                       |
|                         | $>C=C<$ (both C have at least one non-H neighbor)                                                                  | $>C(c)=C(c)<$               | 58, 62   | 2-Heptene, mesityl oxide                       |
|                         | Non-cyclic $>C=C<$ connected to at least one aromatic C                                                            | $>C(c)=C(c)<$ (C(a))        | 59, 59   | Isosafrole, cinnamic alcohol                   |
|                         | Cyclic $>C=C<$                                                                                                     | $>C(r)=C(r)<$               | 62, 60   | Cyclopentadiene                                |
|                         | Non-cyclic $>C=C<$ substituted with at least one F, Cl, N or O                                                     | $-(e)C(c)=C(c)<$            | 60, 58   | Trans-1,2-dichloroethylene, perfluoroisoprene  |
| $-C\equiv C-$           | $HC\equiv C-$ (1-ine)                                                                                              | $HC\equiv C-$               | 64, 56   | 1-Heptyne                                      |
|                         | $-C\equiv C-$ with two non-H neighbors                                                                             | $-C\equiv C-$               | 63, 61   | 2-Octyne                                       |
| $>C=C=C<$               | Two cumulated double bonds                                                                                         | $>C=C=C<$                   | 87, 5    | 1,2-Butadiene, dimethyl allene                 |
| $>C=C-C=C<$             | Two conjugated double bonds in a ring                                                                              | $>C=C-C=C<$                 | 88, 6    | Cyclopentadiene, abietic acid                  |
| $>C=C-C=C<$             | Two conjugated double bonds in a chain                                                                             | $>C=C-C=C<$                 | 89, 7    | Isoprene, 1,3-hexadiene                        |
| $-C\equiv C-C\equiv C-$ | Two conjugated triple bonds                                                                                        | $-C\equiv C-C\equiv C-$     | 95, 8    | 2,4-Hexadiyne                                  |
| Silicon                 |                                                                                                                    |                             |          |                                                |
| $>Si<$                  | $>Si<$                                                                                                             | $>Si<$                      | 70, 80   | Butylsilane                                    |
|                         | $>Si<$ attached to no carbon or hydrogen                                                                           | $>Si<$ (C,H) <sub>0</sub>   | 216, 76  | Tetrachlorosilane, tetramethoxysilane          |
|                         | $>Si<$ attached to one carbon or hydrogen                                                                          | $>Si<$ (C,H) <sub>1</sub>   | 215, 77  | Trichlorosilane, methyltrichlorosilane         |
|                         | $>Si<$ attached to two carbon or hydrogen                                                                          | $>Si<$ (C,H) <sub>2</sub>   | 93, 78   | Dichlorodimethylsilane, dichlorodiethylsilane  |
|                         | $>Si<$ attached to three carbon or hydrogen                                                                        | $>Si<$ (C,H) <sub>3</sub>   | 71, 79   | Hexamethyl disiloxane                          |
| Germanium               |                                                                                                                    |                             |          |                                                |
| $>Ge<$                  | $>Ge<$ connected to four carbons                                                                                   | $(C)_2>Ge<(C)_2$            | 86, 67   | Tetramethylgermane                             |
| $GeCl_3-$               | $GeCl_3-$ connected to carbons                                                                                     | $GeCl_3-$                   | 85, 12   | Fluorodimethylsilyl (trichlorogermanyl)methane |
| Tin                     |                                                                                                                    |                             |          |                                                |
| $>Sn<$                  | $>Sn<$ connected to four carbons                                                                                   | $(C)_2>Sn<(C)_2$            | 83, 64   | Tetramethylstannane                            |
| Periodic Group 13       |                                                                                                                    |                             |          |                                                |
| Boron                   |                                                                                                                    |                             |          |                                                |
| $B(O-)_3$               | Non-cyclic boric acid ester                                                                                        | $B(O-)_3$                   | 78, 15   | Triethyl borate                                |
| Aluminum                |                                                                                                                    |                             |          |                                                |
| $>Al<$                  | $>Al<$ connected to at least one C or Si                                                                           | $>Al<$                      | 117, 47  | Triethylaluminum                               |

Abbreviations: e, very electronegative neighbors (N, O, F, Cl); ne, not very electronegative neighbors (not N, O, F, Cl); na, non-aromatic atom or neighbor; a, aromatic atom or neighbor; c atom or neighbor is part of a chain; r, atom or neighbor is part of a ring; z, groups considered to be non-additive.

<sup>a</sup> Periodic group number of the most significant element in the structural group. Periodic group numbers are used as defined in the IUPAC version of the periodic table of elements.

Table 3  
Second order groups and corrections

| Name                                         | Description                                                                  | ID  | Occurs e.g. in            |
|----------------------------------------------|------------------------------------------------------------------------------|-----|---------------------------|
| C=C–C=O                                      | –C=O connected to sp <sup>2</sup> carbon                                     | 134 | Benzaldehyde, furfural    |
| (C=O)–C([F,Cl] <sub>2,3</sub> )              | Carbonyl connected to C with two or more halogens                            | 119 | Dichloroacetyl chloride   |
| (C=O)–C([F,Cl] <sub>2,3</sub> ) <sub>2</sub> | Carbonyl connected to two C with two or more halogens each                   | 120 | Perfluoro-2-propanone     |
| C–[F,Cl] <sub>3</sub>                        | C with three halogens                                                        | 121 | 1,1,1-Trifluorotoluene    |
| (C) <sub>2</sub> –C–[F,Cl] <sub>2</sub>      | Secondary C with two halogens                                                | 122 | 2,2-Dichloropropane       |
| No hydrogen                                  | Component has no H                                                           | 123 | Perfluoro compounds       |
| One hydrogen                                 | Component has one H                                                          | 124 | Nonafluorobutane          |
| 3/4 Ring                                     | A three or four-membered non-aromatic ring                                   | 125 | Cyclobutene               |
| 5 Ring                                       | A five-membered non-aromatic ring                                            | 126 | Cyclopentane              |
| Ortho pair(s)                                | Ortho position—counted only once and only if there are no meta or para pairs | 127 | <i>o</i> -Xylene          |
| Meta pair(s)                                 | Meta position—counted only once and only if there are no para or ortho pairs | 128 | <i>m</i> -Xylene          |
| Para pair(s)                                 | Para position—counted only once and only if there are no meta or ortho pairs | 129 | <i>p</i> -Xylene          |
| ((C=)(C)C–CC <sub>3</sub> )                  | C–C bond with four single bonded and one double bonded C neighbor            | 130 | <i>tert</i> -Butylbenzene |
| C <sub>2</sub> C–CC <sub>2</sub>             | C–C bond with four C neighbors, two on each side                             | 131 | Bicyclohexyl              |
| C <sub>3</sub> C–CC <sub>2</sub>             | C–C bond with five C neighbors                                               | 132 | Ethyl bornyl ether        |
| C <sub>3</sub> C–CC <sub>3</sub>             | C–C bond with six C neighbors                                                | 133 | 2,2,3,3-Tetramethylbutane |
| Si<(F, Cl, Br, I)                            | Si attached to a halogen atom                                                | 217 | Trichloroethylsilane      |

Table 4  
Groups considered to be non-additive (group–ID(s) given in brackets)

| Group abbr. | Group name (group–ID (s))           | Group abbr.     | Group name (Group–ID (s))                  |
|-------------|-------------------------------------|-----------------|--------------------------------------------|
| OH          | Alkanol (–OH) (34,35,36)            | Ats             | Aromatic sulphur (–S(a)–) (56)             |
| OH(a)       | Phenol (–OH(a)) (37)                | SH              | Thiol (–SH) (53)                           |
| COOH        | Carboxylic acid (–COOH) (44)        | NH <sub>2</sub> | Primary amine (–NH <sub>2</sub> ) (40, 41) |
| EtherO      | Ether (–O–) (38)                    | NH              | Secondary amine (>NH) (42,97)              |
| Epox        | Epoxide (>(OC <sub>2</sub> )<) (39) | OCN             | Isocyanate (–OCN) (80)                     |
| Ester       | Ester (–COOC–) (45,46,47)           | CN              | Cyanide (–CN) (57)                         |
| Ketone      | Ketone (–CO–) (51,92)               | Nitro           | Nitrate (69)                               |
| Alde        | Aldehyde (–CHO) (52,90)             | AN <sub>5</sub> | Aromatic N in 5-ring (=N(a)–(r5)) (66)     |
| AO          | Aromatic oxygen (–O(a)–) (65)       | AN <sub>6</sub> | Aromatic N in 6-ring (=N(a)–(r6)) (67)     |
| Teth        | Sulfide (thioether) (–S(na)–) (54)  |                 |                                            |

Table 5  
Group changes with respect to the previous method

| Periodic group | Structure | Previous name (ID) | New name (ID)                               | Comments                                                                                                         |
|----------------|-----------|--------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 14             | =C(a)<    | New                | C(a)=C(a)<C <sub>2</sub> (a) (bridge) (214) | Aromatic =C< with three aromatic neighbors and two aromatic bonds (aliphatic bridge bond between aromatic rings) |
|                | >Si<      | >Si<(O) (71)       | –Si<(C,H) <sub>3</sub> (71)                 | >Si< attached to no carbon or hydrogen                                                                           |
|                | >Si<      | >Si<(F, Cl) (93)   | >Si<(C,H) <sub>2</sub> (93)                 | >Si< attached to one carbon or hydrogen                                                                          |
|                | >Si<      | New                | >Si<(C,H) (215)                             | >Si< attached to two carbon or hydrogen                                                                          |
|                | >Si<      | New                | >Si< (216)                                  | >Si< attached to three carbon or hydrogen                                                                        |

Group IDs 118 and 134 were switched.

Tables 5 and 6, respectively. An additional group interaction was included for molecules containing both a COOH– and an NH<sub>2</sub>–group. In this case (amino acids), zwitterions<sup>1</sup> are formed and no property estimation can be performed using the new method. The group, correction and interaction contributions are

listed in Tables 7–9 for critical temperature, Tables 10–12 for critical pressure and Tables 13–15 for critical volume, respectively.

For a number of groups listed in Table 2, no parameters are available and critical property data estimation is not possible.

<sup>1</sup> A zwitterion (from German “Zwitter”—“hybrid”, “hermaphrodite”) is a compound with acidic and basic groups in the same molecule. At neutral pH most zwitterions are therefore, negatively charged anions and positively charged cations at the same time. Zwitterions usually have a high solubility in water due to their charged groups and a poor solubility in most organic solvents (<http://www.wikipedia.org>).

Table 6  
Second order groups and corrections changes with respect to the previous method

| Name              | Description                     | ID  | Previously used |
|-------------------|---------------------------------|-----|-----------------|
| Si<(F, Cl, Br, I) | Si attached to any halogen atom | 217 | No              |

Table 7

Critical temperature group contributions, number of components used for regressing these values and deviations for these components

| Group ID | Group contribution<br>( $\times 10^3$ ) | Number of<br>components | Mean absolute<br>error (K) | Mean absolute<br>error (%) | Standard<br>deviation (K) | Group ID | Group contribution<br>( $\times 10^3$ ) | Number of<br>components | Mean absolute<br>error (K) | Mean absolute<br>error (%) | Standard<br>deviation (K) |
|----------|-----------------------------------------|-------------------------|----------------------------|----------------------------|---------------------------|----------|-----------------------------------------|-------------------------|----------------------------|----------------------------|---------------------------|
| 1        | 41.8682                                 | 357                     | 4.5                        | 0.8                        | 6.6                       | 45       | 75.7089                                 | 45                      | 3.3                        | 0.5                        | 4.2                       |
| 2        | 33.1371                                 | 51                      | 3.9                        | 0.7                        | 5.8                       | 46       | 58.0782                                 | 6                       | 2.3                        | 0.4                        | 2.7                       |
| 3        | −1.0710                                 | 42                      | 3.7                        | 0.5                        | 5.1                       | 47       | 109.1930                                | 1                       |                            |                            |                           |
| 4        | 40.0977                                 | 245                     | 4.4                        | 0.7                        | 6.0                       | 48       | 102.1024                                | 2                       | 2.0                        | 0.3                        | 2.0                       |
| 5        | 30.2069                                 | 66                      | 4.2                        | 0.7                        | 6.0                       | 51       | 56.1572                                 | 25                      | 2.7                        | 0.4                        | 3.7                       |
| 6        | −3.8778                                 | 21                      | 2.5                        | 0.4                        | 3.5                       | 52       | 44.2000                                 | 9                       | 2.3                        | 0.4                        | 2.7                       |
| 7        | 52.8003                                 | 231                     | 5.1                        | 0.9                        | 7.3                       | 53       | −7.1070                                 | 7                       | 4.2                        | 0.7                        | 5.6                       |
| 8        | 9.4422                                  | 18                      | 2.2                        | 0.3                        | 3.1                       | 54       | 0.5887                                  | 10                      | 5.5                        | 0.9                        | 9.6                       |
| 9        | 21.2898                                 | 35                      | 3.7                        | 0.6                        | 5.8                       | 56       | −7.7181                                 | 4                       | 1.2                        | 0.2                        | 1.5                       |
| 10       | 26.3513                                 | 11                      | 3.5                        | 0.5                        | 4.5                       | 57       | 117.1330                                | 10                      | 8.3                        | 1.3                        | 10.6                      |
| 11       | −17.0459                                | 3                       | 4.7                        | 0.7                        | 5.6                       | 58       | 45.1531                                 | 4                       | 2.4                        | 0.5                        | 2.9                       |
| 12       | 51.7974                                 | 10                      | 5.5                        | 1.2                        | 6.2                       | 60       | 67.9821                                 | 5                       | 4.0                        | 1.0                        | 4.4                       |
| 13       | 18.9549                                 | 11                      | 4.3                        | 0.8                        | 8.1                       | 61       | 45.4406                                 | 26                      | 5.0                        | 0.9                        | 7.0                       |
| 14       | −29.1568                                | 3                       | 4.6                        | 0.6                        | 5.9                       | 62       | 56.4059                                 | 6                       | 5.7                        | 0.9                        | 5.9                       |
| 15       | 16.1154                                 | 115                     | 3.8                        | 0.5                        | 5.7                       | 63       | −19.9737                                | 1                       |                            |                            |                           |
| 16       | 68.2045                                 | 76                      | 3.5                        | 0.5                        | 5.0                       | 64       | 36.0883                                 | 2                       | 0.2                        | 0.0                        | 0.2                       |
| 17       | 68.1923                                 | 50                      | 3.6                        | 0.5                        | 5.0                       | 65       | 10.4146                                 | 4                       | 1.3                        | 0.2                        | 1.7                       |
| 18       | 29.8039                                 | 12                      | 5.9                        | 0.7                        | 9.7                       | 66       | 18.9903                                 | 2                       | 0.0                        | 0.0                        | 0.0                       |
| 19       | 15.6068                                 | 11                      | 3.1                        | 0.7                        | 3.5                       | 67       | 10.9495                                 | 13                      | 4.6                        | 0.6                        | 7.1                       |
| 20       | 11.0757                                 | 5                       | 3.3                        | 0.9                        | 4.0                       | 68       | 82.6239                                 | 1                       |                            |                            |                           |
| 21       | 18.1302                                 | 56                      | 3.7                        | 0.8                        | 5.1                       | 70       | 25.4209                                 | 2                       | 2.6                        | 0.5                        | 2.8                       |
| 22       | 19.1772                                 | 12                      | 4.4                        | 1.0                        | 6.2                       | 71       | 72.5587                                 | 9                       | 8.6                        | 1.4                        | 12.8                      |
| 23       | 20.8519                                 | 3                       | 2.7                        | 0.7                        | 2.9                       | 76       | 164.3355                                | 1                       |                            |                            |                           |
| 24       | −24.0220                                | 17                      | 2.2                        | 0.4                        | 3.1                       | 78       | 157.3401 <sup>a</sup>                   | 1                       |                            |                            |                           |
| 25       | −1.3329                                 | 14                      | 5.8                        | 1.0                        | 7.1                       | 79       | 97.2830                                 | 1                       |                            |                            |                           |
| 26       | 2.6113                                  | 7                       | 2.8                        | 0.6                        | 3.5                       | 80       | 153.7225                                | 6                       | 8.8                        | 1.2                        | 10.4                      |
| 27       | 15.5010                                 | 22                      | 3.2                        | 0.7                        | 3.9                       | 82       | 90.9726 <sup>b</sup>                    | 1                       |                            |                            |                           |
| 28       | −16.1905                                | 7                       | 4.2                        | 0.7                        | 4.7                       | 83       | 62.3642                                 | 1                       |                            |                            |                           |
| 29       | 60.1907                                 | 3                       | 4.7                        | 1.1                        | 4.9                       | 87       | 53.6350                                 | 1                       |                            |                            |                           |
| 30       | 5.2621                                  | 7                       | 1.4                        | 0.3                        | 1.9                       | 88       | 24.7302                                 | 1                       |                            |                            |                           |
| 31       | −21.5199                                | 2                       | 1.3                        | 0.2                        | 1.3                       | 90       | 38.4681                                 | 1                       |                            |                            |                           |
| 32       | −8.6881                                 | 4                       | 3.9                        | 0.7                        | 4.0                       | 92       | 63.6504                                 | 1                       |                            |                            |                           |
| 33       | 84.8567                                 | 4                       | 8.1                        | 1.5                        | 10.5                      | 93       | 34.2058                                 | 11                      | 9.9                        | 1.5                        | 14.4                      |
| 34       | 79.3047                                 | 27                      | 7.9                        | 1.3                        | 11.1                      | 97       | 27.3441                                 | 3                       | 1.6                        | 0.3                        | 1.9                       |
| 35       | 49.5968                                 | 28                      | 7.4                        | 1.1                        | 9.0                       | 98       | 34.5325                                 | 1                       |                            |                            |                           |
| 36       | 130.1320                                | 16                      | 6.6                        | 1.0                        | 10.7                      | 102      | 1.3231                                  | 4                       | 1.0                        | 0.2                        | 1.3                       |
| 37       | 14.0159                                 | 15                      | 4.0                        | 0.6                        | 4.7                       | 103      | 764.9595 <sup>c</sup>                   | 1                       |                            |                            |                           |
| 38       | 12.5082                                 | 82                      | 6.0                        | 1.1                        | 9.0                       | 110      | 23.4707                                 | 2                       | 3.9                        | 0.6                        | 3.9                       |
| 39       | 41.3490                                 | 3                       | 1.4                        | 0.3                        | 1.7                       | 115      | 36.0361                                 | 1                       |                            |                            |                           |
| 40       | 18.3404                                 | 10                      | 3.0                        | 0.6                        | 4.0                       | 118      | Do not estimate                         |                         |                            |                            |                           |
| 41       | −50.6419                                | 1                       |                            |                            |                           | 214      | 48.1680                                 | 6                       | 6.5                        | 0.8                        | 8.4                       |
| 42       | 17.1780                                 | 7                       | 3.8                        | 0.8                        | 5.6                       | 215      | 0.2842                                  | 5                       | 3.9                        | 0.7                        | 4.0                       |
| 43       | −0.5820                                 | 3                       | 5.1                        | 1.0                        | 6.0                       | 216      | −0.6536                                 | 3                       | 12.6                       | 2.2                        | 12.8                      |
| 44       | 199.9042                                | 13                      | 4.7                        | 0.7                        | 6.1                       |          |                                         |                         |                            |                            |                           |

<sup>a</sup> Based on a publication from 1959, Hansen and Hughes [24].<sup>b</sup> Based on questionable data point by Steele et al. [25].<sup>c</sup> Very questionable, based only on Nesterov et al. [26].

Table 8

Critical temperature second-order contributions, number of components used for regressing these values and deviations for these components

| Group ID | Group contribution ( $\times 10^3$ ) | Number of components | Mean absolute error (K) | Mean absolute error (%) | Standard deviation (K) | Group ID | Group contribution ( $\times 10^3$ ) | Number of components | Mean absolute error (K) | Mean absolute error (%) | Standard deviation (K) |
|----------|--------------------------------------|----------------------|-------------------------|-------------------------|------------------------|----------|--------------------------------------|----------------------|-------------------------|-------------------------|------------------------|
| 119      | 32.1829                              | 2                    | 0.0                     | 0.0                     | 0.0                    | 127      | 1.2823                               | 29                   | 4.1                     | 0.6                     | 5.3                    |
| 120      | 11.4437                              | 2                    | 0.9                     | 0.2                     | 0.9                    | 128      | 6.7099                               | 13                   | 2.5                     | 0.4                     | 3.1                    |
| 121      | −1.3023                              | 56                   | 3.1                     | 0.7                     | 4.1                    | 130      | −33.8201                             | 2                    | 0.0                     | 0.0                     | 0.0                    |
| 122      | −34.3037                             | 21                   | 6.0                     | 1.3                     | 7.3                    | 131      | −18.4815                             | 12                   | 2.9                     | 0.5                     | 3.4                    |
| 123      | −1.3798                              | 44                   | 3.7                     | 0.8                     | 5.0                    | 132      | −23.6024                             | 7                    | 2.8                     | 0.5                     | 3.9                    |
| 124      | −2.7180                              | 17                   | 3.0                     | 0.7                     | 3.6                    | 133      | −24.5802                             | 3                    | 1.4                     | 0.2                     | 1.5                    |
| 125      | 11.3251                              | 9                    | 4.6                     | 1.0                     | 5.6                    | 134      | −35.6113                             | 4                    | 2.0                     | 0.3                     | 3.1                    |
| 126      | −4.7516                              | 18                   | 3.5                     | 0.7                     | 6.5                    | 217      | 62.0286                              | 7                    | 5.9                     | 1.1                     | 7.0                    |

Table 9

Critical temperature group interaction contributions, interacting groups, number of components used for regressing these values and deviations for these components

| Group ID | Interacting groups (see Table 4) | Group contribution ( $\times 10^3$ ) | Number of components | Mean absolute error (K) | Mean absolute error (%) | Standard deviation (K) |
|----------|----------------------------------|--------------------------------------|----------------------|-------------------------|-------------------------|------------------------|
| 135      | OH—OH                            | −434.8568                            | 5                    | 13.1                    | 1.7                     | 17.7                   |
| 136      | OH—NH <sub>2</sub>               | 120.9166                             | 2                    | 0.0                     | 0.0                     | 0.0                    |
| 137      | OH—NH                            | −30.4354                             | 1                    |                         |                         |                        |
| 140      | OH—EtherO                        | −146.7881                            | 10                   | 3.3                     | 0.5                     | 4.2                    |
| 148      | OH(a)—OH(a)                      | 144.4697                             | 1                    |                         |                         |                        |
| 157      | NH <sub>2</sub> —NH <sub>2</sub> | −60.9217                             | 1                    |                         |                         |                        |
| 159      | NH <sub>2</sub> —EtherO          | −738.0515                            | 1                    |                         |                         |                        |
| 165      | NH—NH                            | −49.7641                             | 1                    |                         |                         |                        |
| 176      | OCN—OCN                          | −1866.0970 <sup>a</sup>              | 3                    | 12.8                    | 1.7                     | 13.9                   |
| 178      | EtherO—EtherO                    | 162.6878                             | 31                   | 9.4                     | 1.6                     | 12.8                   |
| 179      | EtherO—Epoxy                     | (707.4116) <sup>b</sup>              | 1                    |                         |                         |                        |
| 180      | EtherO—Ester                     | 128.2740                             | 6                    | 1.5                     | 0.2                     | 1.8                    |
| 183      | EtherO—Teth                      | −654.1363 <sup>c</sup>               | 1                    |                         |                         |                        |
| 185      | EtherO—CN                        | 741.8565                             | 3                    | 13.8                    | 1.8                     | 16.0                   |
| 189      | Ester—Ester                      | 366.2663                             | 2                    | 1.6                     | 0.2                     | 1.7                    |
| 194      | Ketone—Ketone                    | 1605.5640                            | 1                    |                         |                         |                        |
| 205      | Teth—Teth                        | −861.1528 <sup>c</sup>               | 1                    |                         |                         |                        |
| 208      | AtS—AN <sub>5</sub>              | 131.7924                             | 1                    |                         |                         |                        |
| 210      | AO—AN <sub>5</sub>               | 24.0243                              | 1                    |                         |                         |                        |
| 212      | AN <sub>6</sub> —AN <sub>6</sub> | −32.3208                             | 1                    |                         |                         |                        |
| 218      | COOH—NH <sub>2</sub>             | Do not estimate                      | —                    |                         |                         |                        |

<sup>a</sup> This value is of very questionable size. All three data points were reported by Zhuravlev et al. [27] and were not directly experimental.<sup>b</sup> Based on only one questionable publication (Steele et al. [28]).<sup>c</sup> Based on only one questionable publication (Steele et al. [29]).

Table 10  
Critical pressure group contributions, number of components used for regressing these values and deviations for these components

| Group ID | Group contribution<br>( $\times 10^4$ ) | Number of<br>components | Mean absolute<br>error (kPa) | Mean absolute<br>error (%) | Standard deviation<br>(kPa) | Group ID         | Group contribution<br>( $\times 10^4$ ) | Number of<br>components | Mean absolute<br>error (kPa) | Mean absolute<br>error (%) | Standard deviation<br>(kPa) |
|----------|-----------------------------------------|-------------------------|------------------------------|----------------------------|-----------------------------|------------------|-----------------------------------------|-------------------------|------------------------------|----------------------------|-----------------------------|
| 1        | 8.1620                                  | 287                     | 91.5                         | 2.8                        | 138.2                       | 44               | 3.9873                                  | 13                      | 148.3                        | 3.9                        | 243.5                       |
| 2        | 5.5262                                  | 40                      | 130.1                        | 3.0                        | 208.1                       | 45               | 4.3592                                  | 27                      | 97.5                         | 3.6                        | 130.2                       |
| 3        | 4.1660                                  | 26                      | 89.8                         | 2.4                        | 115.0                       | 46               | 1.0266                                  | 3                       | 235.8                        | 4.6                        | 277.4                       |
| 4        | 5.2623                                  | 180                     | 83.4                         | 2.7                        | 123.7                       | 47               | 0.4329                                  | 1                       |                              |                            |                             |
| 5        | 2.3009                                  | 48                      | 93.0                         | 2.8                        | 134.8                       | 48               | 0.5172                                  | 1                       |                              |                            |                             |
| 6        | −2.9925                                 | 20                      | 86.0                         | 3.1                        | 119.1                       | 51               | 0.1190                                  | 17                      | 60.2                         | 1.7                        | 89.8                        |
| 7        | 3.4310                                  | 187                     | 111.4                        | 3.4                        | 157.3                       | 52               | −2.3615                                 | 7                       | 102.9                        | 2.9                        | 121.3                       |
| 8        | 2.3665                                  | 20                      | 78.4                         | 2.8                        | 88.8                        | 53               | −9.4154                                 | 2                       | 411.0                        | 6.4                        | 415.5                       |
| 9        | 3.4027                                  | 28                      | 76.7                         | 1.8                        | 127.2                       | 54               | −8.2595                                 | 5                       | 89.8                         | 2.3                        | 128.2                       |
| 10       | 3.6162                                  | 6                       | 28.3                         | 0.9                        | 36.9                        | 56               | −4.9259                                 | 4                       | 57.7                         | 1.3                        | 70.8                        |
| 11       | −5.1299                                 | 1                       |                              |                            |                             | 57               | 5.1666                                  | 6                       | 157.0                        | 4.3                        | 172.7                       |
| 12       | 4.1421                                  | 11                      | 141.6                        | 5.3                        | 151.9                       | 58               | 7.1581                                  | 4                       | 53.8                         | 1.4                        | 60.6                        |
| 13       | 0.8765                                  | 10                      | 109.0                        | 2.5                        | 175.1                       | 60               | −6.2791                                 | 5                       | 198.0                        | 5.2                        | 238.2                       |
| 14       | −0.1320                                 | 3                       | 15.2                         | 0.4                        | 20.4                        | 61               | 9.6413                                  | 21                      | 112.0                        | 3.3                        | 137.7                       |
| 15       | 2.1064                                  | 87                      | 84.5                         | 2.4                        | 122.8                       | 62               | 3.4731                                  | 3                       | 75.3                         | 1.6                        | 82.4                        |
| 16       | 4.1826                                  | 59                      | 76.8                         | 2.3                        | 105.6                       | 63               | −2.2718                                 | 1                       |                              |                            |                             |
| 17       | 3.5500                                  | 27                      | 83.5                         | 2.0                        | 134.5                       | 64               | 2.4489                                  | 2                       | 157.2                        | 3.1                        | 158.3                       |
| 18       | 1.0997                                  | 9                       | 55.8                         | 1.5                        | 70.4                        | 65               | −0.5403                                 | 4                       | 48.3                         | 1.1                        | 65.1                        |
| 19       | 0.7328                                  | 17                      | 122.9                        | 4.8                        | 144.8                       | 66               | 8.3052                                  | 2                       | 0.0                          | 0.0                        | 0.0                         |
| 20       | 4.3757                                  | 5                       | 218.1                        | 5.1                        | 248.8                       | 67               | −4.7101                                 | 4                       | 89.6                         | 1.7                        | 117.5                       |
| 21       | 3.4933                                  | 72                      | 111.2                        | 4.4                        | 145.3                       | 68               | −5.0929                                 | 1                       |                              |                            |                             |
| 22       | 2.6558                                  | 15                      | 101.4                        | 3.8                        | 132.1                       | 70               | 5.7270                                  | 2                       | 147.4                        | 5.7                        | 168.7                       |
| 23       | 1.6547                                  | 3                       | 153.3                        | 3.0                        | 176.1                       | 71               | 2.7602                                  | 9                       | 59.9                         | 4.0                        | 95.7                        |
| 24       | 0.5236                                  | 12                      | 61.8                         | 1.7                        | 72.7                        | 76               | 4.0458                                  | 1                       |                              |                            |                             |
| 25       | −2.2611                                 | 5                       | 195.7                        | 3.8                        | 234.9                       | 78 <sup>a</sup>  | 12.6786                                 | 1                       |                              |                            |                             |
| 26       | −1.4992                                 | 7                       | 160.4                        | 3.8                        | 189.2                       | 79               | 0.2822                                  | 1                       |                              |                            |                             |
| 27       | 0.4883                                  | 25                      | 74.1                         | 1.8                        | 118.6                       | 82               | −23.9221                                | 1                       |                              |                            |                             |
| 28       | −0.9280                                 | 3                       | 24.7                         | 0.7                        | 25.5                        | 83               | 0.7043                                  | 1                       |                              |                            |                             |
| 29       | 11.8687                                 | 3                       | 195.5                        | 5.1                        | 213.0                       | 87               | 12.6128                                 | 1                       |                              |                            |                             |
| 30       | −4.3170                                 | 6                       | 235.9                        | 5.3                        | 268.0                       | 88               | −10.2451                                | 1                       |                              |                            |                             |
| 31       | −2.2409                                 | 2                       | 140.9                        | 3.7                        | 145.2                       | 90               | −4.0133                                 | 1                       |                              |                            |                             |
| 32       | −4.7841                                 | 2                       | 115.6                        | 2.7                        | 116.0                       | 92               | −5.0403                                 | 1                       |                              |                            |                             |
| 33       | −7.4244                                 | 3                       | 130.1                        | 3.5                        | 154.3                       | 93               | 3.2023                                  | 10                      | 51.6                         | 3.3                        | 68.2                        |
| 34       | −4.4735                                 | 17                      | 78.3                         | 2.0                        | 108.4                       | 97               | −4.3834                                 | 2                       | 0.0                          | 0.0                        | 0.0                         |
| 35       | −1.8153                                 | 28                      | 63.3                         | 2.3                        | 113.8                       | 98               | −13.6078                                | 1                       |                              |                            |                             |
| 36       | −6.8991                                 | 14                      | 169.6                        | 2.9                        | 222.6                       | 102              | 3.3971                                  | 5                       | 28.8                         | 0.7                        | 38.3                        |
| 37       | −12.1664                                | 6                       | 168.2                        | 3.3                        | 219.2                       | 103 <sup>b</sup> | 58.9190                                 | 1                       |                              |                            |                             |
| 38       | 2.0592                                  | 72                      | 111.1                        | 3.7                        | 153.1                       | 110              | 0.1812                                  | 2                       | 37.2                         | 1.1                        | 37.3                        |
| 39       | 0.1759                                  | 2                       | 0.0                          | 0.0                        | 0.0                         | 115              | −5.1116                                 | 1                       |                              |                            |                             |
| 40       | −4.4164                                 | 10                      | 222.2                        | 4.5                        | 338.4                       | 118              | Do not estimate                         | –                       | –                            | –                          | –                           |
| 41       | −9.0065                                 | 1                       |                              |                            |                             | 214              | 1.5574                                  | 6                       | 172.8                        | 6.0                        | 213.2                       |
| 42       | −0.4086                                 | 7                       | 146.5                        | 5.6                        | 202.0                       | 215              | 3.8751                                  | 4                       | 62.0                         | 2.2                        | 82.2                        |
| 43       | 2.3625                                  | 5                       | 88.1                         | 2.4                        | 148.3                       | 216              | 4.4882                                  | 5                       | 39.1                         | 1.2                        | 48.4                        |

<sup>a</sup> Based on a publication from 1959 (Hansen and Hughes [24]).

<sup>b</sup> Very questionable, based on only one publication (Nesterov et al. [26]).

The groups are nevertheless reported. In case data or reasonable estimates for other components containing these groups are available to the user, it is possible to calculate or regress the missing contribution. If the groups would have been discarded from the method, it might still have been possible to perform property estimation using more simple (lower priority) groups, but it would have led to large errors. For interacting groups, not every interaction is of similar importance. Ignoring specific group interactions will lead to similar errors as demonstrated in case of the normal boiling point estimation [21].

### 3.1. Development of the model equations

An important feature of a group contribution method, beside the correlation of available data, is its ability to safely interpolate and extrapolate. This is especially true for critical property data estimation as most available data are for rather small and simple molecules. The methods are frequently applied to very different substances, even to polymers with molecular weights that are orders of magnitude larger than the components in the experimental database.

When analyzing the results for 10 different estimation methods from literature, it became apparent that five of these produced physically unrealistic extrapolations. Fig. 1 shows the estimated critical temperature for *n*-alkanes as function of the number of carbon atoms. As this erroneous behavior is caused by the functional form of the model equation, similar unrealistic extrapolations can be expected for all types of large molecules.

In this work, the following equations were used due to their correlative power and safe extrapolation ability:

$$T_c = T_b \left( b + \frac{1}{a + (\sum_i N_i C_i + \text{GI})^c} \right) \quad (1)$$

$$\frac{P_c}{\text{kPa}} = \frac{M^b (\text{g/mol})}{(a + \sum_i N_i C_i + \text{GI})^2} \quad (2)$$

$$\frac{V_c}{10^{-6} \text{ m}^3 \text{ mol}^{-1}} = \frac{\sum_i N_i C_i + \text{GI}}{n^a} + b \quad (3)$$

with  $N_i$ , number of groups of type  $i$ ;  $C_i$ , group contribution of group  $i$ ;  $M$ , molecular weight (g/mol);  $a$ ,  $b$ ,  $c$ , adjustable parameters (see Table 16);  $n$ , number of atoms in the molecule (except hydrogen); GI, total group interaction contribution (see Eq. (4)).

The model parameters for each property are listed in Table 16.

The model for  $T_c$  is similar to the model used by Ambrose (AB) [5,6], with the exception that the parameter  $b$  replaces the value of unity employed by Ambrose. A value of one entails that  $T_c \rightarrow T_b$  with increasing number of group contribution increments. However, since it is considered possible for  $T_c$  to have values smaller than  $T_b$  for very large compounds (e.g., polymers), the variable  $b$  was regressed to experimental data and a value of  $b$  lower than one was accepted. Several authors have employed a quadratic function (Joback, etc.) in their models which, in all cases, results in a physically unrealistic local maximum or minimum (see Fig. 1). In the case of  $P_c$ , a quadratic function was employed in this work but it was verified, that the

Table 11  
Critical pressure second-order contributions, number of components used for regressing these values and deviations for these components

| Group ID | Group contribution<br>( $\times 10^4$ ) | Number of<br>components | Mean absolute<br>error (kPa) | Mean absolute<br>error (%) | Standard deviation<br>(kPa) | Group ID | Group contribution<br>( $\times 10^4$ ) | Number of<br>components | Mean absolute<br>error (kPa) | Mean absolute<br>error (%) | Standard deviation<br>(kPa) |
|----------|-----------------------------------------|-------------------------|------------------------------|----------------------------|-----------------------------|----------|-----------------------------------------|-------------------------|------------------------------|----------------------------|-----------------------------|
| 119      | 7.3149                                  | 4                       | 162.3                        | 7.5                        | 204.2                       | 127      | 0.7061                                  | 15                      | 121.4                        | 3.8                        | 152.3                       |
| 120      | 4.1439                                  | 2                       | 25.5                         | 0.9                        | 25.5                        | 128      | -0.7246                                 | 8                       | 112.8                        | 3.4                        | 123.8                       |
| 121      | 0.4387                                  | 70                      | 115.0                        | 4.1                        | 154.8                       | 130      | -8.8457                                 | 3                       | 73.1                         | 2.6                        | 100.2                       |
| 122      | -4.2678                                 | 28                      | 138.4                        | 6.6                        | 174.2                       | 131      | -2.2542                                 | 11                      | 66.5                         | 3.1                        | 85.4                        |
| 123      | 4.8944                                  | 51                      | 102.2                        | 4.0                        | 132.5                       | 132      | -3.2460                                 | 5                       | 55.6                         | 2.0                        | 58.5                        |
| 124      | 2.8103                                  | 21                      | 149.8                        | 4.4                        | 184.0                       | 133      | -5.3113                                 | 3                       | 58.3                         | 2.1                        | 63.8                        |
| 125      | -0.3035                                 | 6                       | 79.3                         | 2.5                        | 115.1                       | 134      | 1.0934                                  | 7                       | 58.9                         | 2.7                        | 76.4                        |
| 126      | 0.0930                                  | 18                      | 89.5                         | 2.8                        | 145.4                       | 217      | 8.6126                                  | 10                      | 58.2                         | 1.7                        | 76.6                        |

Table 12

Critical pressure group interaction contributions, interacting groups, number of components used for regressing these values and deviations for these components

| Group ID | Interacting groups<br>(see Table 4) | Group contribution<br>( $\times 10^4$ ) | Number of<br>components | Mean absolute<br>error (kPa) | Mean absolute<br>error (%) | Standard<br>deviation (kPa) |
|----------|-------------------------------------|-----------------------------------------|-------------------------|------------------------------|----------------------------|-----------------------------|
| 135      | OH–OH                               | –5.6023                                 | 5                       | 268.4                        | 4.0                        | 303.5                       |
| 136      | OH–NH <sub>2</sub>                  | 69.8200                                 | 2                       | 0.0                          | 0.0                        | 0.0                         |
| 137      | OH–NH                               | 6.1331                                  | 1                       |                              |                            |                             |
| 140      | OH–EtherO                           | 7.3373                                  | 10                      | 83.3                         | 2.3                        | 126.9                       |
| 148      | OH(a)–OH(a)                         | 57.8350 <sup>a</sup>                    | 1                       |                              |                            |                             |
| 157      | NH <sub>2</sub> –NH <sub>2</sub>    | –0.6754                                 | 1                       |                              |                            |                             |
| 159      | NH <sub>2</sub> –EtherO             | –125.5983                               | 1                       |                              |                            |                             |
| 165      | NH–NH                               | 22.1871                                 | 1                       |                              |                            |                             |
| 178      | EtherO–EtherO                       | 2.6751                                  | 21                      | 69.7                         | 3.0                        | 96.2                        |
| 179      | EtherO–Epox                         | 88.8752 <sup>b</sup>                    | 1                       |                              |                            |                             |
| 180      | EtherO–Ester                        | –1.0295                                 | 5                       | 91.1                         | 3.3                        | 108.7                       |
| 183      | EtherO–Teth                         | 25.8246 <sup>c</sup>                    | 1                       |                              |                            |                             |
| 189      | Ester–Ester                         | 0.5195                                  | 2                       | 30.5                         | 1.0                        | 30.5                        |
| 194      | Ketone–Ketone                       | –78.2743                                | 1                       |                              |                            |                             |
| 205      | Teth–Teth                           | 43.9001 <sup>b</sup>                    | 1                       |                              |                            |                             |
| 208      | AtS–AN <sub>5</sub>                 | –19.7033                                | 1                       |                              |                            |                             |
| 210      | AO–AN <sub>5</sub>                  | –35.1998                                | 1                       |                              |                            |                             |
| 212      | AN <sub>6</sub> –AN <sub>6</sub>    | 12.5371                                 | 1                       |                              |                            |                             |
| 218      | COOH–NH <sub>2</sub>                | Do not estimate                         | –                       |                              |                            |                             |

<sup>a</sup> Based on only one questionable publication by Steele et al. [30].<sup>b</sup> Based on only one questionable publication by Steele et al. [28].<sup>c</sup> Based on only one questionable publication by Steele et al. [29].

location of the maximum does not lead to problems in extrapolation. For  $V_c$ , there is a relatively small dependence on the number of atoms as is reflected by the small negative value of  $a$ .

### 3.2. Hydrocarbon compounds

As with the previous work, model development started with regression of group contributions for saturated, unsaturated and aromatic hydrocarbons. For fused aromatic hydrocarbons, it

became apparent that a new group had to be introduced for compounds where a non-aromatic bond connects two aromatic carbon atoms. This modification was also successfully introduced into our method for normal boiling temperature estimation and will be published together with further revisions and extensions. The results and quality analysis for hydrocarbons are presented in Tables 17–19.

Except for cyclic alkenes and alkynes, the results for  $T_c$  and  $P_c$  are comparable to the method of Ambrose while all other

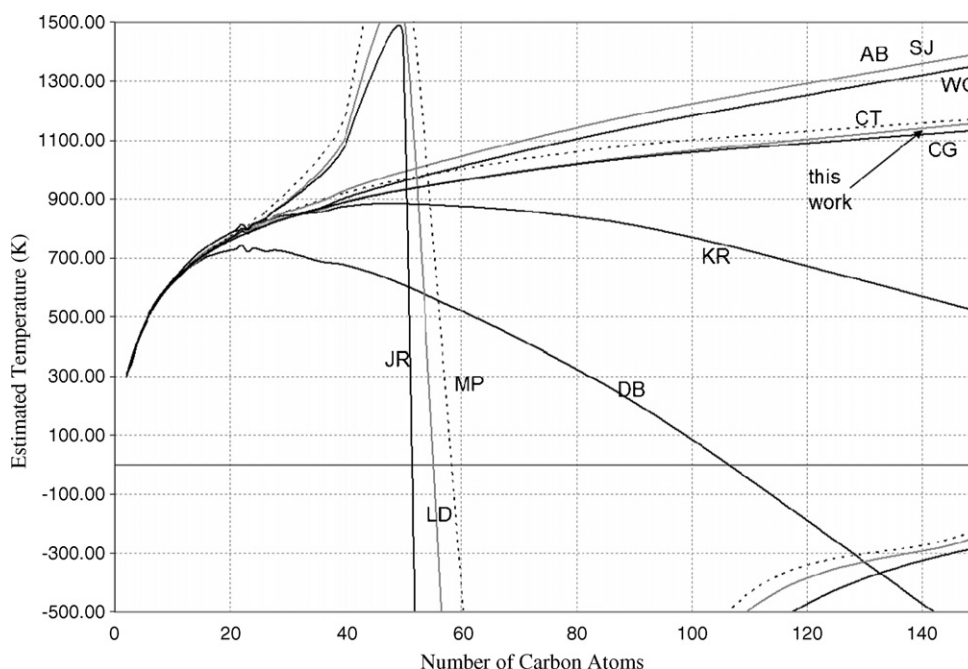


Fig. 1. Estimated critical temperature for  $n$ -alkanes as function of the number of carbon atoms from different group contribution methods (AB and SJ overlap).

Table 13  
Critical volume group contributions, number of components used for regressing these values and deviations for these components

| Group ID | Group contribution | Number of components | Mean absolute error (cm <sup>3</sup> /mol) | Mean absolute error (%) | Standard deviation (cm <sup>3</sup> /mol) | Group ID | Group contribution | Number of components | Mean absolute error (cm <sup>3</sup> /mol) | Mean absolute error (%) | Standard deviation (cm <sup>3</sup> /mol) |
|----------|--------------------|----------------------|--------------------------------------------|-------------------------|-------------------------------------------|----------|--------------------|----------------------|--------------------------------------------|-------------------------|-------------------------------------------|
| 1        | 28.7855            | 196                  | 6.2                                        | 1.5                     | 9.1                                       | 40       | 13.2571            | 4                    | 9.1                                        | 3.3                     | 10.0                                      |
| 2        | 28.8811            | 24                   | 6.3                                        | 2.4                     | 9.6                                       | 41       | 73.7444            | 1                    |                                            |                         |                                           |
| 3        | 26.7237            | 13                   | 5.2                                        | 1.5                     | 7.2                                       | 42       | 20.5722            | 3                    | 3.9                                        | 0.8                     | 4.9                                       |
| 4        | 32.0493            | 136                  | 6.2                                        | 1.4                     | 9.1                                       | 43       | 6.0178             | 2                    | 4.3                                        | 1.0                     | 4.9                                       |
| 5        | 32.1108            | 29                   | 8.7                                        | 2.1                     | 11.7                                      | 44       | 40.3909            | 5                    | 9.8                                        | 3.2                     | 13.5                                      |
| 6        | 28.0534            | 12                   | 9.7                                        | 2.4                     | 12.2                                      | 45       | 42.6733            | 12                   | 5.1                                        | 1.3                     | 8.8                                       |
| 7        | 33.7577            | 122                  | 6.5                                        | 1.9                     | 9.9                                       | 46       | 36.1286            | 3                    | 2.2                                        | 1.0                     | 2.4                                       |
| 8        | 28.8792            | 7                    | 1.5                                        | 0.3                     | 2.5                                       | 48       | 64.3506            | 1                    |                                            |                         |                                           |
| 9        | 24.8517            | 23                   | 7.5                                        | 2.0                     | 11.1                                      | 51       | 30.9229            | 26                   | 4.1                                        | 0.8                     | 6.0                                       |
| 10       | 30.9323            | 8                    | 12.9                                       | 2.9                     | 16.5                                      | 52       | 25.5034            | 4                    | 1.2                                        | 0.6                     | 1.3                                       |
| 11       | 5.9550             | 3                    | 15.0                                       | 3.1                     | 17.3                                      | 53       | 34.7699            | 3                    | 11.5                                       | 4.9                     | 13.8                                      |
| 12       | 29.5901            | 7                    | 8.7                                        | 2.2                     | 11.0                                      | 54       | 38.0185            | 5                    | 2.3                                        | 0.7                     | 3.3                                       |
| 13       | 20.2325            | 7                    | 5.0                                        | 1.9                     | 6.9                                       | 56       | 20.3127            | 4                    | 5.1                                        | 1.7                     | 6.8                                       |
| 14       | 10.5669            | 3                    | 6.6                                        | 1.4                     | 8.4                                       | 57       | 43.7983            | 1                    |                                            |                         |                                           |
| 15       | 19.4020            | 58                   | 4.8                                        | 1.4                     | 7.6                                       | 60       | 51.0710            | 5                    | 4.4                                        | 2.0                     | 5.2                                       |
| 16       | 25.0434            | 31                   | 3.2                                        | 0.8                     | 5.5                                       | 61       | 48.1957            | 15                   | 6.9                                        | 2.2                     | 10.4                                      |
| 17       | 5.6704             | 21                   | 7.8                                        | 2.2                     | 10.9                                      | 62       | 34.1240            | 6                    | 15.1                                       | 3.8                     | 18.1                                      |
| 18       | 16.4118            | 8                    | 3.8                                        | 0.9                     | 5.6                                       | 63       | 40.9263            | 1                    |                                            |                         |                                           |
| 19       | −5.0331            | 11                   | 6.2                                        | 1.9                     | 7.9                                       | 64       | 29.8612            | 2                    | 2.2                                        | 1.2                     | 2.2                                       |
| 20       | 1.5646             | 5                    | 4.8                                        | 2.8                     | 5.5                                       | 65       | 4.7476             | 4                    | 6.8                                        | 2.9                     | 9.1                                       |
| 21       | 3.3646             | 46                   | 6.9                                        | 2.2                     | 10.4                                      | 66       | −25.3680           | 2                    | 0.0                                        | 0.0                     | 0.0                                       |
| 22       | 1.0897             | 9                    | 10.8                                       | 4.8                     | 16.2                                      | 67       | 23.6094            | 5                    | 7.1                                        | 2.4                     | 9.6                                       |
| 23       | 1.1084             | 3                    | 2.6                                        | 1.6                     | 2.6                                       | 68       | 34.8472            | 1                    |                                            |                         |                                           |
| 24       | 19.3190            | 10                   | 7.9                                        | 2.2                     | 9.4                                       | 70       | 75.7193            | 1                    |                                            |                         |                                           |
| 25       | 22.0457            | 4                    | 14.1                                       | 4.3                     | 15.7                                      | 71       | 69.5645            | 6                    | 17.0                                       | 1.5                     | 21.1                                      |
| 26       | 23.9279            | 5                    | 5.0                                        | 1.5                     | 6.1                                       | 79       | 52.8789            | 1                    |                                            |                         |                                           |
| 27       | 26.2582            | 20                   | 4.4                                        | 1.6                     | 5.9                                       | 80       | 27.1026            | 9                    | 17.5                                       | 3.8                     | 21.1                                      |
| 28       | 36.7624            | 7                    | 11.8                                       | 3.2                     | 13.8                                      | 82       | 68.0701            | 1                    |                                            |                         |                                           |
| 29       | 34.4110            | 3                    | 4.1                                        | 1.7                     | 4.8                                       | 88       | 64.4616            | 1                    |                                            |                         |                                           |
| 30       | 36.0223            | 11                   | 5.5                                        | 2.0                     | 6.7                                       | 90       | 20.0440            | 1                    |                                            |                         |                                           |
| 31       | 30.7004            | 1                    |                                            |                         |                                           | 92       | 28.7127            | 1                    |                                            |                         |                                           |
| 32       | 48.2989            | 1                    |                                            |                         |                                           | 93       | 55.3822            | 6                    | 14.9                                       | 1.4                     | 18.6                                      |
| 33       | 10.6790            | 1                    |                                            |                         |                                           | 97       | 29.3068            | 2                    | 0.0                                        | 0.0                     | 0.0                                       |
| 34       | 5.6645             | 20                   | 5.0                                        | 1.1                     | 6.2                                       | 102      | 1.3597             | 4                    | 1.7                                        | 0.6                     | 2.3                                       |
| 35       | 2.0869             | 10                   | 4.8                                        | 0.9                     | 6.3                                       | 115      | 16.2688            | 1                    |                                            |                         |                                           |
| 36       | 3.7778             | 7                    | 2.7                                        | 1.2                     | 3.3                                       | 118      | Do not estimate    | –                    |                                            |                         |                                           |
| 37       | 25.6584            | 2                    | 0.0                                        | 0.0                     | 0.0                                       | 214      | 16.3122            | 6                    | 5.3                                        | 0.9                     | 6.3                                       |
| 38       | 11.6284            | 41                   | 11.6                                       | 3.2                     | 15.3                                      | 215      | 37.0423            | 4                    | 14.6                                       | 2.7                     | 19.0                                      |
| 39       | 46.7680            | 2                    | 0.0                                        | 0.0                     | 0.0                                       | 216      | 55.7432            | 6                    | 10.4                                       | 2.7                     | 11.3                                      |

Table 14  
 Critical volume second-order contributions, number of components used for regressing these values and deviations for these components

| Group ID | Group contribution | Number of components | Mean absolute error (cm <sup>3</sup> /mol) | Mean absolute error (%) | Standard deviation (cm <sup>3</sup> /mol) | Group ID | Group contribution | Number of components | Mean absolute error (cm <sup>3</sup> /mol) | Mean absolute error (%) | Standard deviation (cm <sup>3</sup> /mol) |
|----------|--------------------|----------------------|--------------------------------------------|-------------------------|-------------------------------------------|----------|--------------------|----------------------|--------------------------------------------|-------------------------|-------------------------------------------|
| 119      | −3.8033            | 2                    | 0.0                                        | 0.0                     | 0.0                                       | 127      | −3.5964            | 11                   | 4.4                                        | 1.2                     | 6.7                                       |
| 120      | 27.5326            | 1                    |                                            |                         |                                           | 128      | 1.5196             | 5                    | 3.3                                        | 0.7                     | 4.6                                       |
| 121      | 1.5807             | 50                   | 6.3                                        | 2.2                     | 9.8                                       | 130      | −4.6483            | 2                    | 0.0                                        | 0.0                     | 0.0                                       |
| 122      | −2.6235            | 16                   | 9.1                                        | 2.1                     | 11.1                                      | 131      | −5.0563            | 10                   | 9.5                                        | 2.1                     | 12.6                                      |
| 123      | −5.3091            | 42                   | 6.9                                        | 1.9                     | 9.1                                       | 132      | −6.3267            | 5                    | 12.9                                       | 2.8                     | 15.7                                      |
| 124      | −6.1909            | 18                   | 7.3                                        | 3.2                     | 11.9                                      | 133      | 4.9392             | 1                    |                                            |                         |                                           |
| 125      | 3.2219             | 6                    | 8.9                                        | 2.2                     | 12.8                                      | 134      | 2.8889             | 4                    | 0.0                                        | 0.0                     | 0.0                                       |
| 126      | −6.3900            | 11                   | 4.8                                        | 1.5                     | 7.2                                       | 217      | 19.4348            | 9                    | 8.9                                        | 2.5                     | 9.9                                       |

Table 15  
 Critical volume group interaction contributions, interacting groups, number of components used for regressing these values and deviations for these components

| Group ID | Interacting groups (see Table 4) | Group contribution     | Number of components | Mean absolute error (cm <sup>3</sup> /mol) | Mean absolute error (%) | Standard deviation (cm <sup>3</sup> /mol) |
|----------|----------------------------------|------------------------|----------------------|--------------------------------------------|-------------------------|-------------------------------------------|
| 137      | OH—NH                            | −8.0423                | 1                    |                                            |                         |                                           |
| 140      | OH—EtherO                        | 19.7707                | 4                    | 3.0                                        | 0.9                     | 3.5                                       |
| 148      | OH(a)—OH(a)                      | 97.5425                | 1                    |                                            |                         |                                           |
| 165      | NH—NH                            | −57.1233               | 1                    |                                            |                         |                                           |
| 176      | OCN—OCN                          | 44.1062                | 4                    | 21.8                                       | 3.9                     | 25.7                                      |
| 178      | EtherO—EtherO                    | −23.6366               | 12                   | 15.7                                       | 3.0                     | 19.2                                      |
| 179      | EtherO—Epox                      | −329.5074 <sup>a</sup> | 1                    |                                            |                         |                                           |
| 180      | EtherO—Ester                     | −55.5112               | 2                    | 20.9                                       | 4.7                     | 21.0                                      |
| 183      | EtherO—Teth                      | −37.2468               | 1                    |                                            |                         |                                           |
| 189      | Ester—Ester                      | −74.8680               | 1                    |                                            |                         |                                           |
| 194      | Ketone—Ketone                    | −413.3976              | 1                    |                                            |                         |                                           |
| 205      | Teth—Teth                        | −403.1196 <sup>b</sup> | 1                    |                                            |                         |                                           |
| 208      | AtS—AN <sub>5</sub>              | 164.2930               | 1                    |                                            |                         |                                           |
| 210      | AO—AN <sub>5</sub>               | 217.9243               | 1                    |                                            |                         |                                           |
| 212      | AN <sub>6</sub> —AN <sub>6</sub> | −26.4556               | 1                    |                                            |                         |                                           |
| 218      | COOH—NH <sub>2</sub>             | Do not estimate        | —                    |                                            |                         |                                           |

<sup>a</sup> Based on only one questionable publication by Steele et al. [28].

<sup>b</sup> Based on only one questionable publication by Steele et al. [29].

Table 16  
Parameters values for the Eqs. (1)–(3)

| Property | <i>a</i> | <i>b</i> | <i>c</i> |
|----------|----------|----------|----------|
| $T_c$    | 0.9889   | 0.6990   | 0.8607   |
| $P_c$    | 0.00939  | −0.14041 | –        |
| $V_c$    | −0.2266  | 86.1539  | –        |

methods perform significantly worse in many cases. This is astonishing as the Ambrose method is the second oldest (1978, 1979) and a highly regarded work in this comparison, and it should be expected that it would have served as a reference for all later developments.

It should be noted, that in these and all further tables, the measure of performance of different methods is influenced by the database used. As the training set used in this work differs from that of other authors, small differences are of little importance. In the case of the deviations for cyclic alkanes shown in Table 17, models with deviations in the range of 2–4 K all perform equally well and the differences are insignificant. However, five of the 11 methods show much higher deviations and are not suitable for these types of components.

Tables 20–22 list results for components where steric corrections were used to account for cumulated branching and isomer effects. These types of components can be described nearly equally well using the steric corrections (this work) and

the delta Platt number employed in the Ambrose method as both approaches describe the same structural effects in a similar manner. All other methods give larger errors for at least one of the component groups.

### 3.3. Mono-functional compounds

The group contribution approach is based on the additivity of group increments with respect to the estimated property. In the case of hydrogen bonding or otherwise associating groups, the assumption of simple additivity is not valid. This was observed during the development of the normal boiling temperature estimation method [21] and was also validated in the case of critical properties. Both critical properties and normal boiling temperature are influenced by the same set of molecular properties (energetic interaction, size and form). For this reason it was assumed that the same group contribution scheme as developed for normal boiling temperatures could be used in this work. This has the advantage that the required differentiation can be derived from and validated using a much larger set of experimental data. If this hypothesis holds, then one data point of good quality is sufficient to calculate a group contribution increment.

Model development therefore proceeded by regressing data for compounds that contain only one non-additive (associating) group. A list of these groups can be found in Table 4. Due to

Table 17  
Critical temperature average absolute error (K) of the models for different types of hydrocarbons (number of components given as superscript)

|                      | This work          | JR                 | CG                  | MP                 | AB                 | CT                  | WQ                | DB                  | LD                 | SJ                 | KR                 |
|----------------------|--------------------|--------------------|---------------------|--------------------|--------------------|---------------------|-------------------|---------------------|--------------------|--------------------|--------------------|
| Hydrocarbons (HC)    | 3.5 <sup>141</sup> | 5.7 <sup>141</sup> | 10.3 <sup>142</sup> | 7.1 <sup>123</sup> | 3.1 <sup>134</sup> | 13.6 <sup>143</sup> | 5 <sup>124</sup>  | 14.3 <sup>140</sup> | 6.4 <sup>141</sup> | 4.9 <sup>141</sup> | 7.4 <sup>141</sup> |
| Saturated HC         | 3.1 <sup>78</sup>  | 5.3 <sup>78</sup>  | 7.6 <sup>80</sup>   | 7.8 <sup>78</sup>  | 2.5 <sup>78</sup>  | 8.5 <sup>80</sup>   | 3.6 <sup>78</sup> | 16.7 <sup>78</sup>  | 6.3 <sup>78</sup>  | 5.5 <sup>78</sup>  | 5.8 <sup>78</sup>  |
| Non-aromatic HC      | 3.3 <sup>104</sup> | 5.2 <sup>104</sup> | 7.5 <sup>105</sup>  | 7.0 <sup>96</sup>  | 2.8 <sup>103</sup> | 8.0 <sup>106</sup>  | 3.8 <sup>89</sup> | 14.7 <sup>104</sup> | 6.3 <sup>104</sup> | 4.9 <sup>104</sup> | 5.7 <sup>104</sup> |
| Unsaturated HC       | 3.8 <sup>26</sup>  | 5.0 <sup>26</sup>  | 7.1 <sup>25</sup>   | 3.8 <sup>18</sup>  | 3.8 <sup>25</sup>  | 6.5 <sup>26</sup>   | 4.9 <sup>11</sup> | 8.6 <sup>26</sup>   | 6.3 <sup>26</sup>  | 3.2 <sup>26</sup>  | 5.5 <sup>29</sup>  |
| <i>n</i> -Alkanes    | 4.4 <sup>25</sup>  | 11.2 <sup>25</sup> | 5.9 <sup>27</sup>   | 17.5 <sup>25</sup> | 4.2 <sup>25</sup>  | 4.9 <sup>27</sup>   | 2.3 <sup>25</sup> | 30.8 <sup>25</sup>  | 13.1 <sup>25</sup> | 4.2 <sup>25</sup>  | 11.6 <sup>25</sup> |
| Alkanes (non-cyclic) | 3.3 <sup>65</sup>  | 5.7 <sup>65</sup>  | 6.4 <sup>67</sup>   | 7.6 <sup>65</sup>  | 2.3 <sup>65</sup>  | 7.1 <sup>67</sup>   | 2.4 <sup>65</sup> | 13.1 <sup>65</sup>  | 6.8 <sup>65</sup>  | 5.8 <sup>65</sup>  | 6.3 <sup>65</sup>  |
| Alkanes (cyclic)     | 2.1 <sup>13</sup>  | 3.4 <sup>13</sup>  | 14.0 <sup>13</sup>  | 8.6 <sup>13</sup>  | 3.6 <sup>13</sup>  | 15.9 <sup>13</sup>  | 9.9 <sup>13</sup> | 34.8 <sup>13</sup>  | 3.9 <sup>13</sup>  | 3.9 <sup>13</sup>  | 3.4 <sup>13</sup>  |
| Aromatic HC          | 4.1 <sup>37</sup>  | 6.9 <sup>37</sup>  | 18.0 <sup>37</sup>  | 7.1 <sup>27</sup>  | 3.8 <sup>31</sup>  | 29.4 <sup>37</sup>  | 8.0 <sup>35</sup> | 13.1 <sup>36</sup>  | 6.5 <sup>37</sup>  | 4.7 <sup>37</sup>  | 12.6 <sup>36</sup> |
| Fused aromatic HC    | 6.8 <sup>5</sup>   | 8.8 <sup>5</sup>   | 13.3 <sup>5</sup>   |                    | 2.4 <sup>2</sup>   | 49.1 <sup>5</sup>   | 6.9 <sup>5</sup>  | 12.6 <sup>5</sup>   | 8.1 <sup>5</sup>   | 8.2 <sup>5</sup>   | 9.5 <sup>5</sup>   |
| Alkenes (HC)         | 4.5 <sup>22</sup>  | 4.5 <sup>22</sup>  | 7.1 <sup>22</sup>   | 3.1 <sup>14</sup>  | 2.7 <sup>22</sup>  | 5.8 <sup>22</sup>   | 4.3 <sup>7</sup>  | 9.0 <sup>22</sup>   | 5.6 <sup>22</sup>  | 3.0 <sup>22</sup>  | 5.8 <sup>22</sup>  |
| Alkenes (cyclic) HC  | 6.4 <sup>4</sup>   | 7.6 <sup>4</sup>   | 12.8 <sup>4</sup>   | 5.9 <sup>2</sup>   | 3.3 <sup>4</sup>   | 6.9 <sup>4</sup>    | 5.5 <sup>3</sup>  | 37.3 <sup>4</sup>   | 6.4 <sup>4</sup>   | 5.6 <sup>4</sup>   | 9.7 <sup>4</sup>   |
| Alkynes (HC)         | 0.1 <sup>3</sup>   | 9.3 <sup>3</sup>   | 7.4 <sup>3</sup>    | 7.0 <sup>3</sup>   | 12.4 <sup>3</sup>  | 10.1 <sup>3</sup>   | 7.1 <sup>3</sup>  | 7.4 <sup>3</sup>    | 11.6 <sup>3</sup>  | 4.7 <sup>3</sup>   | 3.4 <sup>3</sup>   |

Table 18  
Critical pressure average absolute error (kPa) of the models for different types of hydrocarbon compounds (number of components given as superscript)

|                      | This work         | JR                 | CG                 | MP                 | AB                | WQ                 | DB                 | LD                 | SJ                 | KR                 |
|----------------------|-------------------|--------------------|--------------------|--------------------|-------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Hydrocarbons (HC)    | 72 <sup>140</sup> | 104 <sup>140</sup> | 139 <sup>139</sup> | 213 <sup>130</sup> | 71 <sup>132</sup> | 106 <sup>128</sup> | 143 <sup>129</sup> | 124 <sup>140</sup> | 127 <sup>140</sup> | 188 <sup>140</sup> |
| Saturated HC         | 56 <sup>80</sup>  | 97 <sup>80</sup>   | 90 <sup>80</sup>   | 134 <sup>80</sup>  | 54 <sup>78</sup>  | 80 <sup>80</sup>   | 80 <sup>77</sup>   | 104 <sup>80</sup>  | 146 <sup>80</sup>  | 129 <sup>80</sup>  |
| Non-aromatic HC      | 59 <sup>99</sup>  | 95 <sup>99</sup>   | 93 <sup>98</sup>   | 230 <sup>98</sup>  | 62 <sup>96</sup>  | 86 <sup>89</sup>   | 86 <sup>94</sup>   | 108 <sup>99</sup>  | 133 <sup>99</sup>  | 131 <sup>99</sup>  |
| Unsaturated HC       | 75 <sup>19</sup>  | 86 <sup>19</sup>   | 108 <sup>18</sup>  | 653 <sup>18</sup>  | 97 <sup>18</sup>  | 137 <sup>9</sup>   | 113 <sup>17</sup>  | 123 <sup>19</sup>  | 77 <sup>19</sup>   | 138 <sup>19</sup>  |
| <i>n</i> -Alkanes    | 43 <sup>27</sup>  | 72 <sup>27</sup>   | 63 <sup>27</sup>   | 68 <sup>27</sup>   | 65 <sup>25</sup>  | 40 <sup>27</sup>   | 69 <sup>25</sup>   | 71 <sup>27</sup>   | 71 <sup>27</sup>   | 225 <sup>27</sup>  |
| Alkanes (non-cyclic) | 59 <sup>68</sup>  | 100 <sup>68</sup>  | 78 <sup>68</sup>   | 72 <sup>68</sup>   | 39 <sup>66</sup>  | 70 <sup>68</sup>   | 58 <sup>65</sup>   | 99 <sup>68</sup>   | 153 <sup>68</sup>  | 137 <sup>68</sup>  |
| Alkanes (cyclic)     | 39 <sup>12</sup>  | 80 <sup>12</sup>   | 160 <sup>12</sup>  | 486 <sup>12</sup>  | 138 <sup>12</sup> | 134 <sup>12</sup>  | 198 <sup>12</sup>  | 136 <sup>12</sup>  | 107 <sup>12</sup>  | 86 <sup>12</sup>   |
| Aromatic HC          | 103 <sup>41</sup> | 126 <sup>41</sup>  | 248 <sup>41</sup>  | 164 <sup>32</sup>  | 92 <sup>36</sup>  | 152 <sup>39</sup>  | 297 <sup>35</sup>  | 164 <sup>41</sup>  | 113 <sup>41</sup>  | 326 <sup>41</sup>  |
| Fused aromatic HC    | 60 <sup>4</sup>   | 156 <sup>4</sup>   | 382 <sup>4</sup>   |                    | 18 <sup>2</sup>   | 245 <sup>4</sup>   | 604 <sup>4</sup>   | 318 <sup>4</sup>   | 34 <sup>4</sup>    | 108 <sup>4</sup>   |
| Alkenes (HC)         | 74 <sup>15</sup>  | 80 <sup>15</sup>   | 108 <sup>15</sup>  | 804 <sup>14</sup>  | 93 <sup>15</sup>  | 125 <sup>5</sup>   | 90 <sup>15</sup>   | 67 <sup>15</sup>   | 71 <sup>15</sup>   | 134 <sup>15</sup>  |
| Alkenes (cyclic) HC  | 33 <sup>1</sup>   | 38 <sup>1</sup>    | 432 <sup>1</sup>   | 272 <sup>1</sup>   | 434 <sup>1</sup>  | 163 <sup>1</sup>   | 370 <sup>1</sup>   | 8 <sup>1</sup>     | 12 <sup>1</sup>    | 252 <sup>1</sup>   |
| Alkynes (HC)         | 105 <sup>3</sup>  | 142 <sup>3</sup>   | 104 <sup>3</sup>   | 87 <sup>3</sup>    | 115 <sup>3</sup>  | 129 <sup>3</sup>   | 285 <sup>2</sup>   | 248 <sup>3</sup>   | 97 <sup>3</sup>    | 148 <sup>3</sup>   |

Table 19

Critical volume average relative error (%) of the models for different types of hydrocarbon compounds (number of components given as superscript)

|                      | This work         | JR                | CG                | MP                | AB                | WQ                | LD                | SJ                | KR                |
|----------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Hydrocarbons (HC)    | 1.6 <sup>96</sup> | 2.4 <sup>96</sup> | 4.3 <sup>96</sup> | 2.4 <sup>83</sup> | 2.9 <sup>92</sup> | 3.9 <sup>84</sup> | 2.8 <sup>96</sup> | 3.1 <sup>96</sup> | 3.1 <sup>96</sup> |
| Saturated HC         | 1.7 <sup>57</sup> | 2.8 <sup>57</sup> | 3.3 <sup>57</sup> | 2.7 <sup>57</sup> | 3.0 <sup>57</sup> | 3.5 <sup>57</sup> | 3.0 <sup>57</sup> | 3.6 <sup>57</sup> | 3.1 <sup>57</sup> |
| Non-aromatic HC      | 1.9 <sup>74</sup> | 2.6 <sup>74</sup> | 3.1 <sup>74</sup> | 2.4 <sup>70</sup> | 2.8 <sup>74</sup> | 4.2 <sup>63</sup> | 2.8 <sup>74</sup> | 3.3 <sup>74</sup> | 2.9 <sup>74</sup> |
| Unsaturated HC       | 2.5 <sup>17</sup> | 2.0 <sup>17</sup> | 2.5 <sup>17</sup> | 1.2 <sup>13</sup> | 2.3 <sup>17</sup> | 11.0 <sup>6</sup> | 2.3 <sup>17</sup> | 2.6 <sup>17</sup> | 2.4 <sup>17</sup> |
| <i>n</i> -Alkanes    | 1.0 <sup>17</sup> | 5.0 <sup>17</sup> | 5.3 <sup>17</sup> | 4.7 <sup>17</sup> | 5.8 <sup>17</sup> | 4.7 <sup>17</sup> | 5.8 <sup>17</sup> | 5.8 <sup>17</sup> | 6.1 <sup>17</sup> |
| Alkanes (non-cyclic) | 1.8 <sup>49</sup> | 3.1 <sup>49</sup> | 3.1 <sup>49</sup> | 2.6 <sup>49</sup> | 3.3 <sup>49</sup> | 2.9 <sup>49</sup> | 3.3 <sup>49</sup> | 3.9 <sup>49</sup> | 3.5 <sup>49</sup> |
| Alkanes (cyclic)     | 0.9 <sup>8</sup>  | 1.4 <sup>8</sup>  | 5.1 <sup>8</sup>  | 3.6 <sup>8</sup>  | 1.0 <sup>8</sup>  | 7.1 <sup>8</sup>  | 0.9 <sup>8</sup>  | 1.4 <sup>8</sup>  | 0.9 <sup>8</sup>  |
| Aromatic HC          | 0.8 <sup>22</sup> | 1.7 <sup>22</sup> | 8.0 <sup>22</sup> | 2.3 <sup>13</sup> | 3.0 <sup>18</sup> | 2.9 <sup>21</sup> | 2.6 <sup>22</sup> | 2.3 <sup>22</sup> | 3.8 <sup>22</sup> |
| Fused aromatic HC    | 1.2 <sup>3</sup>  | 1.5 <sup>3</sup>  | 9.2 <sup>3</sup>  |                   | 9.0 <sup>2</sup>  | 1.4 <sup>3</sup>  | 1.5 <sup>3</sup>  | 1.5 <sup>3</sup>  | 5.8 <sup>3</sup>  |
| Alkenes (HC)         | 2.9 <sup>14</sup> | 1.8 <sup>14</sup> | 2.6 <sup>14</sup> | 0.9 <sup>10</sup> | 2.3 <sup>14</sup> | 19.4 <sup>3</sup> | 2.1 <sup>14</sup> | 2.7 <sup>14</sup> | 2.3 <sup>14</sup> |
| Alkenes (cyclic HC)  | 4.9 <sup>4</sup>  | 3.4 <sup>4</sup>  | 5.0 <sup>4</sup>  | 1.6 <sup>1</sup>  | 4.3 <sup>4</sup>  | 19.4 <sup>3</sup> | 3.1 <sup>4</sup>  | 4.5 <sup>4</sup>  | 3.6 <sup>4</sup>  |
| Alkynes (HC)         | 0.8 <sup>3</sup>  | 2.6 <sup>3</sup>  | 2.1 <sup>3</sup>  | 2.1 <sup>3</sup>  | 2.4 <sup>3</sup>  | 2.7 <sup>3</sup>  | 3.1 <sup>3</sup>  | 2.3 <sup>3</sup>  | 2.8 <sup>3</sup>  |

Table 20

Critical temperature average absolute error (K) for branched alkanes (number of components given as superscript)

|                                  | This work         | JR                | CG                 | MP                | AB                | CT                 | WQ                | DB                 | LD                | SJ                 | KR                |
|----------------------------------|-------------------|-------------------|--------------------|-------------------|-------------------|--------------------|-------------------|--------------------|-------------------|--------------------|-------------------|
| (C,C=)C–CC <sub>3</sub>          | 0.0 <sup>2</sup>  | 6.6 <sup>2</sup>  | 40.4 <sup>2</sup>  | 14.6 <sup>2</sup> | 3.4 <sup>2</sup>  | 28.4 <sup>1</sup>  | 20.3 <sup>2</sup> | 33.3 <sup>2</sup>  | 27.8 <sup>2</sup> | 3.7 <sup>2</sup>   | 19.7 <sup>2</sup> |
| C <sub>2</sub> C–CC <sub>2</sub> | 2.9 <sup>12</sup> | 3.4 <sup>12</sup> | 16.9 <sup>12</sup> | 5.0 <sup>12</sup> | 2.6 <sup>12</sup> | 13.9 <sup>12</sup> | 6.4 <sup>12</sup> | 15.9 <sup>12</sup> | 3.8 <sup>12</sup> | 12.1 <sup>12</sup> | 5.0 <sup>12</sup> |
| C <sub>3</sub> C–CC <sub>2</sub> | 2.8 <sup>7</sup>  | 4.8 <sup>7</sup>  | 13.2 <sup>7</sup>  | 2.4 <sup>6</sup>  | 1.7 <sup>7</sup>  | 8.2 <sup>7</sup>   | 3.1 <sup>7</sup>  | 13.4 <sup>7</sup>  | 4.5 <sup>7</sup>  | 12.0 <sup>7</sup>  | 6.0 <sup>7</sup>  |
| C <sub>3</sub> C–CC <sub>3</sub> | 1.4 <sup>3</sup>  | 1.5 <sup>3</sup>  | 10.4 <sup>3</sup>  | 1.1 <sup>3</sup>  | 1.7 <sup>3</sup>  | 12.6 <sup>3</sup>  | 2.6 <sup>3</sup>  | 0.9 <sup>3</sup>   | 5.2 <sup>3</sup>  | 16.6 <sup>3</sup>  | 2.8 <sup>3</sup>  |

the lack of experimental data, not all group contribution values could be regressed.

Tables 23–25 contain the results for mono-functional oxygenated components and diols, triols, etc., which were included for convenience. The acronym HC stands for “hydrocarbon” and specifies that the functional group is connected to a hydrocarbon backbone without any addition of further functional groups. The extraordinary good results of the Ambrose method for  $T_c$  and  $P_c$  of alkanols are not matched by any other method. In this work, slightly larger deviations were accepted to improve the results for multifunctional compounds. Due to the additional conjugated bond correction for ketones, aldehydes and carboxylic acids, the new method yields significantly better results for these components.

For the case of halogenated compounds, no group interaction is considered. Therefore, Tables 26–28 contain results for components with any number of halogen atoms. The new method

is applicable to the largest variety of components and produces the best overall results. For sulphur and nitrogen compounds the quality results are presented in Tables 29–31 and 32–34 respectively.

The quality analysis for various other elemental groups are presented in Tables 35–37. In the case of the regression of silicon compounds, the groups were entirely re-engineered regarding the number of electronegative neighbours of a silicon atom. A new correction was also developed for silicon atoms that do not have any halogen atoms attached. The modification was also applied to our previous method [21] with similar success.

### 3.4. Multi-functional compounds

As in the previous method, a group interaction contribution for non-additive groups is adopted in this study. The total group

Table 21

Critical pressure average absolute error (kPa) branched alkanes (number of components given as superscript)

|                                  | This work        | JR                | CG                | MP               | AB                | WQ               | DB                | LD                | SJ                | KR               |
|----------------------------------|------------------|-------------------|-------------------|------------------|-------------------|------------------|-------------------|-------------------|-------------------|------------------|
| (C,C=)C–CC <sub>3</sub>          | 73 <sup>3</sup>  | 281 <sup>3</sup>  | 250 <sup>3</sup>  | 168 <sup>3</sup> | 116 <sup>3</sup>  | 164 <sup>3</sup> | 278 <sup>2</sup>  | 236 <sup>3</sup>  | 200 <sup>3</sup>  | 194 <sup>3</sup> |
| C <sub>2</sub> C–CC <sub>2</sub> | 67 <sup>11</sup> | 120 <sup>11</sup> | 301 <sup>11</sup> | 41 <sup>11</sup> | 222 <sup>10</sup> | 58 <sup>11</sup> | 250 <sup>10</sup> | 163 <sup>11</sup> | 346 <sup>11</sup> | 53 <sup>11</sup> |
| C <sub>3</sub> C–CC <sub>2</sub> | 56 <sup>5</sup>  | 191 <sup>5</sup>  | 105 <sup>5</sup>  | 42 <sup>5</sup>  | 14 <sup>5</sup>   | 84 <sup>5</sup>  | 51 <sup>5</sup>   | 215 <sup>5</sup>  | 409 <sup>5</sup>  | 83 <sup>5</sup>  |
| C <sub>3</sub> C–CC <sub>3</sub> | 58 <sup>3</sup>  | 304 <sup>3</sup>  | 68 <sup>3</sup>   | 58 <sup>3</sup>  | 17 <sup>3</sup>   | 109 <sup>3</sup> | 92 <sup>3</sup>   | 342 <sup>3</sup>  | 510 <sup>3</sup>  | 180 <sup>3</sup> |

Table 22

Critical volume average relative error (%) for branched alkanes (number of components given as superscript)

|                                  | This work         | JR                | CG               | MP               | AB               | WQ                | LD                | SJ                | KR                |
|----------------------------------|-------------------|-------------------|------------------|------------------|------------------|-------------------|-------------------|-------------------|-------------------|
| (C,C=)C–CC <sub>3</sub>          | 0.0 <sup>2</sup>  | 6.6 <sup>2</sup>  | 8.4 <sup>2</sup> |                  | 9.9 <sup>2</sup> | 6.2 <sup>2</sup>  | 9.2 <sup>2</sup>  | 9.6 <sup>2</sup>  | 6.4 <sup>2</sup>  |
| C <sub>2</sub> C–CC <sub>2</sub> | 2.1 <sup>10</sup> | 3.6 <sup>10</sup> | 1.8 <sup>7</sup> | 3.2 <sup>9</sup> | 1.7 <sup>7</sup> | 8.7 <sup>10</sup> | 3.0 <sup>10</sup> | 5.6 <sup>10</sup> | 3.6 <sup>10</sup> |
| C <sub>3</sub> C–CC <sub>2</sub> | 2.8 <sup>5</sup>  | 4.0 <sup>5</sup>  | 2.3 <sup>5</sup> | 1.6 <sup>3</sup> | 3.3 <sup>5</sup> | 2.8 <sup>5</sup>  | 3.7 <sup>5</sup>  | 6.5 <sup>5</sup>  | 3.6 <sup>5</sup>  |
| C <sub>3</sub> C–CC <sub>3</sub> | 0.0 <sup>1</sup>  | 0.2 <sup>1</sup>  | 8.3 <sup>1</sup> | 1.3 <sup>1</sup> | 3.0 <sup>1</sup> | 0.8 <sup>1</sup>  | 1.9 <sup>1</sup>  | 3.9 <sup>1</sup>  | 0.1 <sup>1</sup>  |

Table 23

Critical temperature average absolute error (K) of the models for different types of oxygenated compounds (number of components given as superscript)

|                           | This work          | JR                 | CG                  | MP                 | AB                 | CT                  | WQ                 | DB                  | LD                  | SJ                  | KR                 |
|---------------------------|--------------------|--------------------|---------------------|--------------------|--------------------|---------------------|--------------------|---------------------|---------------------|---------------------|--------------------|
| 1-Alkanols (HC)           | 7.1 <sup>18</sup>  | 23.0 <sup>18</sup> | 11.4 <sup>19</sup>  | 17.8 <sup>18</sup> | 2.4 <sup>18</sup>  | 5.5 <sup>19</sup>   | 20.5 <sup>18</sup> | 66.1 <sup>18</sup>  | 24.9 <sup>18</sup>  | 4.9 <sup>18</sup>   | 11.1 <sup>18</sup> |
| Alkanols                  | 6.4 <sup>88</sup>  | 21.1 <sup>88</sup> | 18.5 <sup>94</sup>  | 16.9 <sup>88</sup> | 8.1 <sup>88</sup>  | 24.2 <sup>79</sup>  | 17.9 <sup>87</sup> | 44.2 <sup>87</sup>  | 28.0 <sup>88</sup>  | 14.1 <sup>54</sup>  | 13.0 <sup>88</sup> |
| Primary alkanols (HC)     | 7.4 <sup>26</sup>  | 21.0 <sup>26</sup> | 12.0 <sup>27</sup>  | 16.0 <sup>26</sup> | 2.6 <sup>26</sup>  | 7.0 <sup>27</sup>   | 18.0 <sup>25</sup> | 56.0 <sup>25</sup>  | 22.3 <sup>26</sup>  | 5.4 <sup>25</sup>   | 10.0 <sup>26</sup> |
| Secondary alkanols (HC)   | 6.6 <sup>23</sup>  | 18.2 <sup>23</sup> | 8.3 <sup>26</sup>   | 13.3 <sup>23</sup> | 4.6 <sup>23</sup>  | 18.1 <sup>26</sup>  | 11.5 <sup>23</sup> | 14.8 <sup>23</sup>  | 19.3 <sup>23</sup>  |                     | 9.5 <sup>23</sup>  |
| Tertiary alkanols (HC)    | 8.1 <sup>4</sup>   | 11.9 <sup>4</sup>  | 12.3 <sup>4</sup>   | 10.3 <sup>4</sup>  | 7.6 <sup>4</sup>   | 38.6 <sup>4</sup>   | 8.5 <sup>4</sup>   | 13.7 <sup>4</sup>   | 13.0 <sup>4</sup>   |                     | 8.4 <sup>4</sup>   |
| Aromatic alkanols (HC)    | 4.2 <sup>13</sup>  | 4.4 <sup>13</sup>  | 8.9 <sup>13</sup>   | 3.5 <sup>13</sup>  | 4.3 <sup>13</sup>  |                     | 10.2 <sup>13</sup> | 21.1 <sup>13</sup>  | 36.0 <sup>13</sup>  | 3.8 <sup>12</sup>   | 14.5 <sup>13</sup> |
| Diols, triols, etc (HC)   | 13.1 <sup>5</sup>  | 72.9 <sup>5</sup>  | 116.0 <sup>5</sup>  | 62.8 <sup>5</sup>  | 45.6 <sup>5</sup>  | 109.0 <sup>5</sup>  | 55.8 <sup>5</sup>  | 84.6 <sup>5</sup>   | 73.9 <sup>5</sup>   | 60.3 <sup>3</sup>   | 44.7 <sup>5</sup>  |
| Ethers (HC)               | 4.6 <sup>27</sup>  | 3.7 <sup>27</sup>  | 8.8 <sup>23</sup>   | 3.4 <sup>27</sup>  | 4.0 <sup>27</sup>  | 10.8 <sup>24</sup>  | 7.5 <sup>25</sup>  | 29.3 <sup>26</sup>  | 4.5 <sup>27</sup>   | 2.6 <sup>22</sup>   | 4.6 <sup>27</sup>  |
| Epoxides (HC)             | 2.1 <sup>2</sup>   | 2.9 <sup>2</sup>   | 114.0 <sup>1</sup>  | 3.7 <sup>2</sup>   | 7.6 <sup>2</sup>   |                     | 8.7 <sup>2</sup>   | 42.0 <sup>2</sup>   | 4.5 <sup>2</sup>    |                     | 1.2 <sup>2</sup>   |
| Aldehydes (HC, no =O)     | 2.3 <sup>9</sup>   | 11.0 <sup>9</sup>  | 6.2 <sup>8</sup>    | 4.4 <sup>9</sup>   | 5.5 <sup>9</sup>   | 6.8 <sup>9</sup>    | 4.0 <sup>9</sup>   | 37.7 <sup>9</sup>   | 12.7 <sup>9</sup>   | 46.6 <sup>9</sup>   | 4.5 <sup>9</sup>   |
| Aldehydes (HC, =O)        | 0.0 <sup>1</sup>   | 17.6 <sup>1</sup>  | 5.9 <sup>1</sup>    | 4.1 <sup>1</sup>   | 8.4 <sup>1</sup>   | 37.9 <sup>1</sup>   | 3.2 <sup>1</sup>   | 8.7 <sup>1</sup>    | 20.4 <sup>1</sup>   | 2.4 <sup>1</sup>    | 5.9 <sup>1</sup>   |
| Ketones (HC, no =O)       | 2.8 <sup>21</sup>  | 6.3 <sup>21</sup>  | 4.8 <sup>39</sup>   | 5.9 <sup>21</sup>  | 4.0 <sup>19</sup>  | 7.2 <sup>39</sup>   | 4.6 <sup>21</sup>  | 10.1 <sup>21</sup>  | 6.9 <sup>21</sup>   | 11.2 <sup>21</sup>  | 4.4 <sup>19</sup>  |
| Ketones (HC, =O)          | 2.9 <sup>2</sup>   | 0.2 <sup>1</sup>   | 33.1 <sup>1</sup>   |                    | 2.3 <sup>1</sup>   | 51.6 <sup>2</sup>   | 4.9 <sup>1</sup>   | 49.3 <sup>1</sup>   | 1.9 <sup>1</sup>    | 28.6 <sup>1</sup>   | 4.3 <sup>1</sup>   |
| Non-cyclic carbonates     | 0.0 <sup>2</sup>   | 79.5 <sup>2</sup>  | 12.0 <sup>1</sup>   | 51.0 <sup>1</sup>  | 10.6 <sup>1</sup>  |                     |                    |                     | 74.5 <sup>2</sup>   | 9.1 <sup>1</sup>    | 8.3 <sup>1</sup>   |
| Carboxylic acids (HC)     | 5.1 <sup>12</sup>  | 6.5 <sup>12</sup>  | 11.6 <sup>12</sup>  | 4.6 <sup>12</sup>  | 6.9 <sup>12</sup>  | 11.7 <sup>12</sup>  | 5.4 <sup>12</sup>  | 8.6 <sup>12</sup>   | 7.4 <sup>12</sup>   | 7.1 <sup>12</sup>   | 8.5 <sup>12</sup>  |
| Esters (HC)               | 3.7 <sup>37</sup>  | 5.5 <sup>37</sup>  | 7.2 <sup>37</sup>   | 5.4 <sup>36</sup>  | 4.6 <sup>37</sup>  | 17.7 <sup>37</sup>  | 4.0 <sup>36</sup>  | 30.1 <sup>37</sup>  | 4.8 <sup>37</sup>   | 4.9 <sup>37</sup>   | 4.7 <sup>37</sup>  |
| Formic acids esters (HC)  | 2.3 <sup>6</sup>   |                    | 8.8 <sup>6</sup>    | 2.6 <sup>6</sup>   | 3.6 <sup>6</sup>   |                     |                    | 26.4 <sup>6</sup>   | 14.9 <sup>6</sup>   | 1.4 <sup>6</sup>    | 4.5 <sup>6</sup>   |
| Lactones (HC)             | 0.0 <sup>1</sup>   | 0.1 <sup>1</sup>   | 163.0 <sup>1</sup>  |                    |                    |                     |                    | 8.4 <sup>1</sup>    | 8.6 <sup>1</sup>    |                     |                    |
| Anhydrides chains (HC)    | 0.0 <sup>1</sup>   | 2.5 <sup>1</sup>   | 37.1 <sup>1</sup>   |                    | 1.3 <sup>1</sup>   |                     |                    | 7.6 <sup>1</sup>    | 0.5 <sup>1</sup>    | 39.7 <sup>1</sup>   | 6.4 <sup>1</sup>   |
| Aromatic oxygen (HC)      | 1.8 <sup>3</sup>   | 6.5 <sup>3</sup>   |                     | 2.4 <sup>2</sup>   | 10.4 <sup>1</sup>  | 55.0 <sup>3</sup>   | 2.5 <sup>3</sup>   |                     | 5.4 <sup>3</sup>    | 3.0 <sup>3</sup>    | 4.1 <sup>3</sup>   |
| Oxygenated (w/o alkanols) | 4.4 <sup>185</sup> | 8.9 <sup>161</sup> | 14.3 <sup>171</sup> | 6.8 <sup>154</sup> | 5.6 <sup>159</sup> | 23.6 <sup>181</sup> | 7.8 <sup>152</sup> | 37.8 <sup>146</sup> | 10.0 <sup>179</sup> | 10.7 <sup>160</sup> | 7.2 <sup>163</sup> |

interaction contribution can be calculated from the following equation:

$$GI = \frac{1}{n} \sum_{i=1}^m \sum_{j=1}^m \frac{C_{i-j}}{m-1} \quad (\text{where } C_{i-j} = C_{j-i}) \quad (4)$$

with  $C_{i-j}$ , is the group interaction contribution between group  $i$  and group  $j$  [K] ( $C_{i-j}=0$ );  $n$ , the number of atoms (except hydrogen);  $m$ , the total number of interaction groups in the molecule.

The second-order group interactions account for the flawed assumption that all groups are additive. As discussed above (Section 3.3), this assumption is not valid for strongly associating groups exhibiting hydrogen bonding interactions. Increasing the number of these groups leads to competitive association. Although, halogen compounds may act as hydrogen bonding acceptors, halogen groups can be considered to behave additively due to the weakness of these bonds. The same holds for any groups containing  $\pi$ -electrons. This omission does not lead to significantly worse results or any serious estimation failures. The results of the quality analysis for all components exhibiting

Table 24

Critical pressure average absolute error (kPa) of the models for different types of oxygen compounds (number of components given as superscript)

|                           | This work          | JR                 | CG                 | MP                 | AB                 | WQ                 | DB                | LD                 | SJ                 | KR                 |
|---------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------------|--------------------|--------------------|--------------------|
| 1-Alkanols (HC)           | 48 <sup>18</sup>   | 67 <sup>18</sup>   | 159 <sup>18</sup>  | 57 <sup>18</sup>   | 133 <sup>18</sup>  | 72 <sup>18</sup>   | 128 <sup>18</sup> | 73 <sup>18</sup>   | 89 <sup>18</sup>   | 159 <sup>18</sup>  |
| Alkanols                  | 101 <sup>66</sup>  | 258 <sup>66</sup>  | 389 <sup>66</sup>  | 249 <sup>66</sup>  | 345 <sup>64</sup>  | 245 <sup>66</sup>  | 264 <sup>63</sup> | 290 <sup>66</sup>  | 327 <sup>44</sup>  | 305 <sup>66</sup>  |
| Primary alkanols (HC)     | 72 <sup>24</sup>   | 74 <sup>24</sup>   | 164                | 72 <sup>24</sup>   | 137 <sup>24</sup>  | 96 <sup>24</sup>   | 137 <sup>23</sup> | 84 <sup>24</sup>   | 80 <sup>23</sup>   | 170 <sup>24</sup>  |
| Secondary alkanols (HC)   | 63 <sup>13</sup>   | 156 <sup>13</sup>  | 171 <sup>13</sup>  | 198 <sup>13</sup>  | 157 <sup>13</sup>  | 185 <sup>13</sup>  | 128 <sup>13</sup> | 149 <sup>13</sup>  |                    | 247 <sup>13</sup>  |
| Tertiary alkanols (HC)    | 130 <sup>3</sup>   | 266 <sup>3</sup>   | 168 <sup>3</sup>   | 102 <sup>3</sup>   | 255 <sup>3</sup>   | 236 <sup>3</sup>   | 36 <sup>3</sup>   | 274 <sup>3</sup>   |                    | 293 <sup>3</sup>   |
| Aromatic alkanols (HC)    | 230 <sup>4</sup>   | 204 <sup>4</sup>   | 247 <sup>4</sup>   | 227 <sup>4</sup>   | 160 <sup>4</sup>   | 339 <sup>4</sup>   | 280 <sup>4</sup>  | 434 <sup>4</sup>   | 156 <sup>4</sup>   | 170 <sup>4</sup>   |
| Diols, triols, etc (HC)   | 268 <sup>5</sup>   | 865 <sup>5</sup>   | 1963 <sup>5</sup>  | 778 <sup>5</sup>   | 1534 <sup>5</sup>  | 543 <sup>5</sup>   | 935 <sup>5</sup>  | 679 <sup>3</sup>   | 1002 <sup>3</sup>  | 617 <sup>5</sup>   |
| Ethers (HC)               | 157 <sup>23</sup>  | 165 <sup>23</sup>  | 126 <sup>18</sup>  | 125 <sup>23</sup>  | 118 <sup>23</sup>  | 117 <sup>21</sup>  | 164 <sup>19</sup> | 143 <sup>23</sup>  | 108 <sup>18</sup>  | 257 <sup>23</sup>  |
| Epoxides (HC)             | 0 <sup>1</sup>     | 228 <sup>1</sup>   | 498 <sup>1</sup>   | 888 <sup>1</sup>   | 498 <sup>1</sup>   | 780 <sup>1</sup>   | 167 <sup>1</sup>  | 233 <sup>1</sup>   |                    | 714 <sup>1</sup>   |
| Aldehydes (HC, no =O)     | 90 <sup>8</sup>    | 189 <sup>8</sup>   | 229 <sup>8</sup>   | 235 <sup>8</sup>   | 208 <sup>8</sup>   | 171 <sup>8</sup>   |                   | 407 <sup>8</sup>   | 4479 <sup>8</sup>  | 181 <sup>8</sup>   |
| Aldehydes (HC, =O)        | 0 <sup>1</sup>     | 115 <sup>1</sup>   | 293 <sup>1</sup>   | 106 <sup>1</sup>   | 175 <sup>1</sup>   | 73 <sup>1</sup>    |                   | 418 <sup>1</sup>   | 2 <sup>1</sup>     | 610 <sup>1</sup>   |
| Ketones (HC, no =O)       | 65 <sup>15</sup>   | 67 <sup>14</sup>   | 147 <sup>15</sup>  | 147 <sup>13</sup>  | 71 <sup>12</sup>   | 73 <sup>14</sup>   | 228 <sup>14</sup> | 171 <sup>14</sup>  | 151 <sup>14</sup>  | 94 <sup>12</sup>   |
| Ketones (HC, =O)          | 40 <sup>2</sup>    | 5 <sup>1</sup>     | 548 <sup>2</sup>   |                    | 179 <sup>1</sup>   | 50 <sup>1</sup>    | 711 <sup>1</sup>  | 440 <sup>1</sup>   | 480 <sup>1</sup>   | 301 <sup>1</sup>   |
| Non-cyclic carbonates     | 0 <sup>2</sup>     | 1655 <sup>2</sup>  | 327 <sup>1</sup>   | 748 <sup>1</sup>   | 339 <sup>1</sup>   |                    |                   | 1365 <sup>2</sup>  | 342 <sup>1</sup>   | 284 <sup>1</sup>   |
| Carboxylic acids (HC)     | 146 <sup>12</sup>  | 291 <sup>12</sup>  | 463 <sup>12</sup>  | 614 <sup>12</sup>  | 153 <sup>12</sup>  | 333 <sup>12</sup>  | 310 <sup>12</sup> | 350 <sup>12</sup>  | 134 <sup>12</sup>  | 160 <sup>12</sup>  |
| Esters (HC)               | 89 <sup>15</sup>   | 104 <sup>15</sup>  | 126 <sup>15</sup>  | 63 <sup>14</sup>   | 72 <sup>15</sup>   | 51 <sup>14</sup>   | 189 <sup>14</sup> | 97 <sup>15</sup>   | 59 <sup>15</sup>   | 140 <sup>15</sup>  |
| Formic acids esters (HC)  | 236 <sup>3</sup>   |                    | 238 <sup>3</sup>   | 540 <sup>3</sup>   | 186 <sup>3</sup>   |                    | 92 <sup>3</sup>   | 299 <sup>3</sup>   | 186 <sup>3</sup>   | 315 <sup>3</sup>   |
| Lactones (HC)             | 0 <sup>1</sup>     | 268 <sup>1</sup>   | 392 <sup>1</sup>   |                    |                    |                    | 1409 <sup>1</sup> | 429 <sup>1</sup>   |                    |                    |
| Anhydrides chains (HC)    | 0 <sup>1</sup>     | 250 <sup>1</sup>   | 615 <sup>1</sup>   |                    | 430 <sup>1</sup>   |                    | 1016 <sup>1</sup> | 284 <sup>1</sup>   | 680 <sup>1</sup>   | 248 <sup>1</sup>   |
| Aromatic oxygen (HC)      | 64 <sup>1</sup>    | 198 <sup>1</sup>   |                    | 572 <sup>1</sup>   | 468 <sup>1</sup>   | 530 <sup>1</sup>   |                   | 416 <sup>1</sup>   | 204 <sup>1</sup>   | 512 <sup>1</sup>   |
| Oxygenated (w/o alkanols) | 102 <sup>144</sup> | 281 <sup>127</sup> | 242 <sup>106</sup> | 231 <sup>113</sup> | 165 <sup>111</sup> | 209 <sup>115</sup> | 295 <sup>89</sup> | 231 <sup>142</sup> | 480 <sup>124</sup> | 232 <sup>126</sup> |

Table 25

Critical volume average relative error (%) of the models for different types of oxygen compounds (number of components given as superscript)

|                           | This work         | JR                | CG                | MP                | AB                | WQ                | LD                | SJ                | KR                |
|---------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 1-Alkanols (HC)           | 1.2 <sup>10</sup> | 2.5 <sup>10</sup> | 2.5 <sup>10</sup> | 2.7 <sup>10</sup> | 2.6 <sup>10</sup> | 2.5 <sup>10</sup> | 2.8 <sup>10</sup> | 2.7 <sup>10</sup> | 2.7 <sup>10</sup> |
| Alkanols                  | 1.0 <sup>40</sup> | 3.6 <sup>40</sup> | 4.2 <sup>40</sup> | 3.7 <sup>37</sup> | 3.9 <sup>38</sup> | 3.6 <sup>40</sup> | 3.3 <sup>40</sup> | 3.9 <sup>18</sup> | 4.0 <sup>40</sup> |
| Primary alkanols (HC)     | 1.3 <sup>11</sup> | 2.3 <sup>11</sup> | 2.3 <sup>11</sup> | 2.7 <sup>11</sup> | 2.6 <sup>11</sup> | 2.6 <sup>11</sup> | 2.6 <sup>11</sup> | 2.4 <sup>11</sup> | 2.6 <sup>11</sup> |
| Secondary alkanols (HC)   | 1.1 <sup>20</sup> | 2.9 <sup>20</sup> | 3.9 <sup>20</sup> | 2.9 <sup>19</sup> | 3.9 <sup>20</sup> | 3.3 <sup>20</sup> | 3.1 <sup>20</sup> |                   | 3.3 <sup>20</sup> |
| Tertiary alkanols (HC)    | 0.0 <sup>1</sup>  | 2.6 <sup>1</sup>  | 3.4 <sup>1</sup>  | 0.2 <sup>1</sup>  | 5.9 <sup>1</sup>  | 0.2 <sup>1</sup>  | 3.8 <sup>1</sup>  |                   | 3.0 <sup>1</sup>  |
| Ethers (HC)               | 2.5 <sup>15</sup> | 1.9 <sup>15</sup> | 3.3 <sup>11</sup> | 2.1 <sup>14</sup> | 2.5 <sup>15</sup> | 3.5 <sup>14</sup> | 2.2 <sup>15</sup> | 1.5 <sup>10</sup> | 2.6 <sup>15</sup> |
| Epoxides (HC)             | 0.0 <sup>1</sup>  | 6.4 <sup>1</sup>  | 4.7 <sup>1</sup>  | 7.6 <sup>1</sup>  | 1.4 <sup>1</sup>  | 19.8 <sup>1</sup> | 0.3 <sup>1</sup>  |                   | 5.7 <sup>1</sup>  |
| Aldehydes (HC, no =O)     | 0.5 <sup>5</sup>  | 6.8 <sup>5</sup>  | 5.1 <sup>5</sup>  | 5.3 <sup>4</sup>  | 1.1 <sup>5</sup>  | 5.4 <sup>5</sup>  | 7.5 <sup>5</sup>  | 23.7 <sup>5</sup> | 4.4 <sup>5</sup>  |
| Aldehydes (HC, =O)        | 0.0 <sup>1</sup>  | 5.7 <sup>1</sup>  | 11.7 <sup>1</sup> |                   | 0.4 <sup>1</sup>  | 7.1 <sup>1</sup>  | 4.9 <sup>1</sup>  | 0.7 <sup>1</sup>  | 7.2 <sup>1</sup>  |
| Ketones (HC, no =O)       | 0.8 <sup>25</sup> | 4.9 <sup>24</sup> | 6.1 <sup>25</sup> | 4.0 <sup>23</sup> | 5.6 <sup>24</sup> | 4.3 <sup>24</sup> | 5.8 <sup>24</sup> | 5.8 <sup>24</sup> | 5.1 <sup>24</sup> |
| Ketones (HC, =O)          | 0.0 <sup>2</sup>  | 3.0 <sup>1</sup>  | 2.4 <sup>2</sup>  |                   | 2.4 <sup>1</sup>  | 2.7 <sup>1</sup>  | 3.3 <sup>1</sup>  | 0.1 <sup>1</sup>  | 2.7 <sup>1</sup>  |
| Non-cyclic carbonates     | 0.0 <sup>1</sup>  | 2.7 <sup>1</sup>  | 3.2 <sup>1</sup>  |                   | 0.8 <sup>1</sup>  |                   | 0.9 <sup>1</sup>  | 1.4 <sup>1</sup>  | 1.3 <sup>1</sup>  |
| Carboxylic acids (HC)     | 4.0 <sup>4</sup>  | 2.9 <sup>4</sup>  | 3.5 <sup>4</sup>  | 2.6 <sup>4</sup>  | 3.8 <sup>4</sup>  | 2.8 <sup>4</sup>  | 3.7 <sup>4</sup>  | 3.2 <sup>4</sup>  | 3.2 <sup>4</sup>  |
| Esters (HC)               | 0.7 <sup>9</sup>  | 1.2 <sup>9</sup>  | 1.3 <sup>9</sup>  | 1.5 <sup>8</sup>  | 1.6 <sup>9</sup>  | 1.6 <sup>9</sup>  | 1.4 <sup>9</sup>  | 1.4 <sup>9</sup>  | 1.9 <sup>9</sup>  |
| Formic acids esters (HC)  | 1.0 <sup>3</sup>  |                   | 1.6 <sup>3</sup>  | 2.0 <sup>3</sup>  | 0.8 <sup>3</sup>  |                   | 6.7 <sup>3</sup>  | 0.5 <sup>3</sup>  | 1.7 <sup>3</sup>  |
| Aromatic oxygen (HC)      | 3.8 <sup>3</sup>  | 4.3 <sup>3</sup>  |                   | 0.2 <sup>2</sup>  | 1.3 <sup>1</sup>  | 2.0 <sup>3</sup>  | 3.9 <sup>3</sup>  | 4.5 <sup>3</sup>  | 6.3 <sup>3</sup>  |
| Oxygenated (w/o alkanols) | 2.0 <sup>96</sup> | 5.1 <sup>84</sup> | 5.6 <sup>71</sup> | 4.8 <sup>77</sup> | 4.1 <sup>73</sup> | 5.8 <sup>81</sup> | 8.6 <sup>94</sup> | 6.2 <sup>87</sup> | 5.3 <sup>87</sup> |

Table 26

Critical temperature average absolute error (K) of the models for different types of halogen compounds (number of components given as superscript)

|                          | This work          | JR                 | CG                  | MP                | AB                 | CT                  | WQ                 | DB                | LD                 | SJ                 | KR                 |
|--------------------------|--------------------|--------------------|---------------------|-------------------|--------------------|---------------------|--------------------|-------------------|--------------------|--------------------|--------------------|
| Fluorinated saturated HC | 3.1 <sup>5</sup>   | 4.0 <sup>5</sup>   | 48.1 <sup>5</sup>   | 2.2 <sup>5</sup>  | 7.5 <sup>5</sup>   | 22.6 <sup>5</sup>   | 3.6 <sup>4</sup>   | 19.9 <sup>3</sup> | 5.4 <sup>5</sup>   | 6.3 <sup>5</sup>   | 4.5 <sup>5</sup>   |
| Fluorinated HC           | 2.1 <sup>21</sup>  | 7.9 <sup>21</sup>  | 28.1 <sup>25</sup>  | 3.0 <sup>20</sup> | 9.2 <sup>21</sup>  | 28.9 <sup>25</sup>  | 2.4 <sup>18</sup>  | 5.8 <sup>19</sup> | 9.1 <sup>21</sup>  | 2.8 <sup>21</sup>  | 3.2 <sup>21</sup>  |
| Chlorinated saturated HC | 4.5 <sup>14</sup>  | 4.7 <sup>14</sup>  | 6.5 <sup>13</sup>   |                   | 4.9 <sup>14</sup>  | 12.9 <sup>14</sup>  | 4.8 <sup>14</sup>  | 7.8 <sup>13</sup> | 4.5 <sup>14</sup>  | 5.2 <sup>13</sup>  | 6.5 <sup>14</sup>  |
| Chlorinated HC           | 4.5 <sup>17</sup>  | 5.1 <sup>17</sup>  | 6.5 <sup>16</sup>   |                   | 4.9 <sup>17</sup>  | 11.9 <sup>173</sup> | 4.8 <sup>16</sup>  | 7.0 <sup>16</sup> | 5.1 <sup>17</sup>  | 4.7 <sup>16</sup>  | 6.5 <sup>17</sup>  |
| Brominated saturated HC  | 2.6 <sup>2</sup>   | 2.6 <sup>2</sup>   | 3.6 <sup>2</sup>    |                   | 5.7 <sup>2</sup>   | 5.1 <sup>2148</sup> | 7.6 <sup>2</sup>   | 16.0 <sup>2</sup> | 10.4 <sup>2</sup>  | 8.0 <sup>2</sup>   | 1.7 <sup>2</sup>   |
| Brominated HC            | 2.2 <sup>3</sup>   | 3.1 <sup>3</sup>   | 5.6 <sup>3</sup>    |                   | 4.2 <sup>3</sup>   | 5.7 <sup>3</sup>    | 5.3 <sup>3</sup>   | 11.7 <sup>3</sup> | 8.5 <sup>3</sup>   | 5.8 <sup>3</sup>   | 2.2 <sup>3</sup>   |
| Iodinated HC             | 4.2 <sup>3</sup>   | 2.6 <sup>3</sup>   | 6.5 <sup>3</sup>    | 0.1 <sup>3</sup>  | 1.4 <sup>3</sup>   | 11.5                | 0.1 <sup>2</sup>   | 0.2 <sup>2</sup>  | 2.7 <sup>3</sup>   | 3.0 <sup>3</sup>   | 4.1 <sup>3</sup>   |
| Halogenated compounds    | 3.6 <sup>126</sup> | 6.0 <sup>115</sup> | 30.9 <sup>128</sup> | 3.8 <sup>60</sup> | 8.4 <sup>115</sup> | 39.8                | 6.3 <sup>111</sup> | 9.9 <sup>82</sup> | 7.2 <sup>121</sup> | 8.9 <sup>121</sup> | 5.7 <sup>115</sup> |

Table 27

Critical pressure average absolute error (kPa) of the models for different types of halogen compounds (number of components given as superscript)

|                          | This work          | JR                 | CG                | MP                | AB                | WQ                | DB                | LD                 | SJ                 | KR                 |
|--------------------------|--------------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|--------------------|--------------------|--------------------|
| Fluorinated saturated HC | 112 <sup>32</sup>  | 242 <sup>32</sup>  | 318 <sup>32</sup> | 122 <sup>32</sup> | 524 <sup>27</sup> | 163 <sup>31</sup> | 367 <sup>20</sup> | 304 <sup>32</sup>  | 511 <sup>32</sup>  | 249 <sup>32</sup>  |
| Fluorinated HC           | 109 <sup>43</sup>  | 244 <sup>43</sup>  | 269 <sup>43</sup> | 107 <sup>42</sup> | 496 <sup>38</sup> | 136 <sup>40</sup> | 296 <sup>31</sup> | 320 <sup>43</sup>  | 415 <sup>43</sup>  | 211 <sup>43</sup>  |
| Chlorinated saturated HC | 211 <sup>9</sup>   | 585 <sup>9</sup>   | 200 <sup>6</sup>  |                   | 533 <sup>9</sup>  | 290 <sup>9</sup>  | 606 <sup>5</sup>  | 275 <sup>9</sup>   | 173 <sup>8</sup>   | 416 <sup>9</sup>   |
| Chlorinated HC           | 183 <sup>11</sup>  | 593 <sup>11</sup>  | 261 <sup>8</sup>  |                   | 477 <sup>11</sup> | 324 <sup>11</sup> | 576 <sup>6</sup>  | 279 <sup>11</sup>  | 155 <sup>9</sup>   | 483 <sup>11</sup>  |
| Brominated saturated HC  | 354 <sup>2</sup>   | 238 <sup>2</sup>   | 753 <sup>2</sup>  |                   | 978 <sup>2</sup>  | 172 <sup>2</sup>  | 1532 <sup>2</sup> | 1026 <sup>2</sup>  | 263 <sup>2</sup>   | 236 <sup>2</sup>   |
| Brominated HC            | 295 <sup>3</sup>   | 367 <sup>3</sup>   | 530 <sup>3</sup>  |                   | 653 <sup>3</sup>  | 195 <sup>3</sup>  | 1082 <sup>3</sup> | 882 <sup>3</sup>   | 225 <sup>3</sup>   | 392 <sup>3</sup>   |
| Iodinated HC             | 125 <sup>1</sup>   | 14 <sup>1</sup>    | 130 <sup>1</sup>  | 14 <sup>1</sup>   | 48 <sup>1</sup>   | 1 <sup>1</sup>    | 7 <sup>1</sup>    | 193 <sup>1</sup>   | 235 <sup>1</sup>   | 11 <sup>1</sup>    |
| Halogenated compounds    | 119 <sup>122</sup> | 395 <sup>112</sup> | 330 <sup>91</sup> | 131 <sup>63</sup> | 559 <sup>91</sup> | 179 <sup>99</sup> | 355 <sup>60</sup> | 304 <sup>121</sup> | 331 <sup>116</sup> | 287 <sup>112</sup> |

Table 28

Critical volume average relative error (%) of the models for different types of halogen compounds (number of components given as superscript)

|                          | This work         | JR                | CG                | MP                | AB                | WQ                | LD                | SJ                | KR                |
|--------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Fluorinated saturated HC | 1.9 <sup>26</sup> | 4.2 <sup>26</sup> | 6.1 <sup>21</sup> | 3.0 <sup>25</sup> | 19.0 <sup>3</sup> | 5.7 <sup>25</sup> | 4.6 <sup>26</sup> | 3.8 <sup>26</sup> | 4.2 <sup>26</sup> |
| Fluorinated HC           | 2.1 <sup>36</sup> | 4.9 <sup>36</sup> | 5.8 <sup>31</sup> | 3.0 <sup>34</sup> | 8.3 <sup>8</sup>  | 4.9 <sup>33</sup> | 4.5 <sup>36</sup> | 3.3 <sup>36</sup> | 3.5 <sup>36</sup> |
| Chlorinated saturated HC | 3.8 <sup>4</sup>  | 4.0 <sup>4</sup>  | 2.3 <sup>2</sup>  |                   | 6.5 <sup>3</sup>  | 4.4 <sup>4</sup>  | 4.1 <sup>4</sup>  | 4.8 <sup>3</sup>  | 3.6 <sup>4</sup>  |
| Chlorinated HC           | 3.0 <sup>6</sup>  | 4.7 <sup>6</sup>  | 4.1 <sup>4</sup>  |                   | 5.0 <sup>4</sup>  | 5.5 <sup>6</sup>  | 5.2 <sup>6</sup>  | 3.7 <sup>4</sup>  | 5.1 <sup>6</sup>  |
| Brominated saturated HC  | 2.1 <sup>1</sup>  | 2.5 <sup>1</sup>  | 2.6 <sup>1</sup>  |                   | 1.1 <sup>1</sup>  | 0.0 <sup>1</sup>  | 2.4 <sup>1</sup>  | 0.8 <sup>1</sup>  | 0.2 <sup>1</sup>  |
| Brominated HC            | 1.1 <sup>2</sup>  | 1.6 <sup>2</sup>  | 3.4 <sup>2</sup>  |                   | 1.4 <sup>2</sup>  | 0.1 <sup>2</sup>  | 2.3 <sup>2</sup>  | 0.6 <sup>2</sup>  | 3.0 <sup>2</sup>  |
| Iodinated HC             | 0.0 <sup>1</sup>  | 0.2 <sup>1</sup>  | 4.7 <sup>1</sup>  | 0.1 <sup>1</sup>  | 0.3 <sup>1</sup>  | 0.0 <sup>1</sup>  | 1.5 <sup>1</sup>  | 0.0 <sup>1</sup>  | 0.2 <sup>1</sup>  |
| Halogenated Compounds    | 2.3 <sup>94</sup> | 5.0 <sup>80</sup> | 5.1 <sup>61</sup> | 5.3 <sup>45</sup> | 7.6 <sup>18</sup> | 5.0 <sup>74</sup> | 6.3 <sup>87</sup> | 4.4 <sup>89</sup> | 4.4 <sup>80</sup> |

Table 29

Critical temperature average absolute error (K) of the models for different types of sulphur compounds (number of components given as superscript)

|                         | This work         | JR                | CG                 | MP                | AB                | CT                 | WQ                | DB                 | LD                | SJ                | KR                 |
|-------------------------|-------------------|-------------------|--------------------|-------------------|-------------------|--------------------|-------------------|--------------------|-------------------|-------------------|--------------------|
| Thiols (HC)             | 4.2 <sup>7</sup>  | 3.0 <sup>7</sup>  | 9.5 <sup>5</sup>   | 0.7 <sup>6</sup>  | 4.4 <sup>7</sup>  | 11.8 <sup>7</sup>  | 2.6 <sup>6</sup>  | 10.7 <sup>6</sup>  | 4.7 <sup>7</sup>  | 1.4 <sup>6</sup>  | 9.0 <sup>7</sup>   |
| Thioether (HC)          | 7.7 <sup>7</sup>  | 9.1 <sup>7</sup>  | 15.5 <sup>7</sup>  | 7.7 <sup>7</sup>  | 9.4 <sup>7</sup>  | 15.4 <sup>6</sup>  | 4.5 <sup>6</sup>  | 24.9 <sup>7</sup>  | 10.1 <sup>7</sup> | 10.2 <sup>6</sup> | 12.8 <sup>7</sup>  |
| Aromatic thioether (HC) | 1.6 <sup>3</sup>  | 9.2 <sup>3</sup>  |                    | 0.5 <sup>1</sup>  |                   | 52.5 <sup>3</sup>  | 4.9 <sup>3</sup>  |                    | 8.3 <sup>3</sup>  | 3.7 <sup>3</sup>  | 2.6 <sup>3</sup>   |
| Sulfolane (O=S=O) (HC)  | 0.0 <sup>1</sup>  |                   |                    |                   |                   |                    | 30.4 <sup>1</sup> |                    |                   |                   |                    |
| Sulphur compounds       | 4.1 <sup>22</sup> | 5.9 <sup>21</sup> | 13.0 <sup>12</sup> | 4.2 <sup>14</sup> | 6.8 <sup>15</sup> | 29.4 <sup>20</sup> | 8.1 <sup>19</sup> | 18.4 <sup>13</sup> | 7.2 <sup>20</sup> | 6.1 <sup>19</sup> | 11.0 <sup>21</sup> |

Table 30

Critical pressure average absolute error (kPa) of the models for different types of sulphur compounds (number of components given as superscript)

|                         | This work         | JR                | CG               | MP               | AB               | WQ                | DB               | LD                | SJ                | KR                |
|-------------------------|-------------------|-------------------|------------------|------------------|------------------|-------------------|------------------|-------------------|-------------------|-------------------|
| Thiols (HC)             | 411 <sup>2</sup>  | 431 <sup>2</sup>  | 240 <sup>1</sup> | 14 <sup>2</sup>  | 181 <sup>2</sup> | 210 <sup>2</sup>  | 641 <sup>2</sup> | 173 <sup>2</sup>  | 185 <sup>2</sup>  | 280 <sup>2</sup>  |
| Thioether (HC)          | 150 <sup>3</sup>  | 219 <sup>3</sup>  | 97 <sup>3</sup>  | 177 <sup>3</sup> | 106 <sup>3</sup> | 178 <sup>3</sup>  | 158 <sup>2</sup> | 101 <sup>3</sup>  | 112 <sup>3</sup>  | 113 <sup>3</sup>  |
| Aromatic thioether (HC) | 77 <sup>3</sup>   | 98 <sup>3</sup>   |                  | 10 <sup>1</sup>  |                  | 338 <sup>3</sup>  |                  | 526 <sup>3</sup>  | 352 <sup>3</sup>  | 497 <sup>3</sup>  |
| Sulfolane (O=S=O) (HC)  | 0 <sup>1</sup>    |                   |                  |                  |                  | 1963 <sup>1</sup> |                  |                   |                   |                   |
| Sulphur compounds       | 125 <sup>12</sup> | 338 <sup>11</sup> | 132 <sup>4</sup> | 95 <sup>6</sup>  | 136 <sup>5</sup> | 652 <sup>12</sup> | 399 <sup>4</sup> | 259 <sup>10</sup> | 237 <sup>11</sup> | 303 <sup>11</sup> |

Table 31

Critical volume average relative error (%) of the models for different types of sulphur compounds (number of components given as superscript)

|                         | This work         | JR                | CG                | MP               | AB               | WQ                 | LD                | SJ                | KR                |
|-------------------------|-------------------|-------------------|-------------------|------------------|------------------|--------------------|-------------------|-------------------|-------------------|
| Thiols (HC)             | 4.9 <sup>3</sup>  | 4.8 <sup>3</sup>  | 16.5 <sup>1</sup> | 0.8 <sup>2</sup> | 4.1 <sup>3</sup> | 0.7 <sup>2</sup>   | 4.2 <sup>3</sup>  | 1.6 <sup>2</sup>  | 3.4 <sup>3</sup>  |
| Thioether (HC)          | 1.2 <sup>3</sup>  | 3.1 <sup>3</sup>  | 2.7 <sup>3</sup>  | 2.5 <sup>3</sup> | 2.9 <sup>3</sup> | 2.6 <sup>3</sup>   | 2.9 <sup>3</sup>  | 2.9 <sup>3</sup>  | 2.7 <sup>3</sup>  |
| Aromatic thioether (HC) | 2.3 <sup>3</sup>  | 1.3 <sup>3</sup>  |                   | 0.0 <sup>1</sup> |                  | 28.5 <sup>3</sup>  | 3.0 <sup>3</sup>  | 1.3 <sup>3</sup>  | 3.0 <sup>3</sup>  |
| Sulfolane (O=S=O) (HC)  | 0.0 <sup>1</sup>  |                   |                   |                  |                  | 58.4 <sup>1</sup>  |                   |                   |                   |
| Sulphur compounds       | 1.9 <sup>13</sup> | 3.5 <sup>12</sup> | 6.1 <sup>4</sup>  | 1.5 <sup>6</sup> | 3.5 <sup>6</sup> | 19.1 <sup>12</sup> | 3.4 <sup>11</sup> | 2.8 <sup>11</sup> | 2.7 <sup>12</sup> |

Table 32

Critical temperature average absolute error (K) of the models for different types of nitrogen compounds (number of components given as superscript)

|                            | This work         | JR                | CG                 | MP                | AB                | CT                 | WQ                | DB                 | LD                 | SJ                  | KR                |
|----------------------------|-------------------|-------------------|--------------------|-------------------|-------------------|--------------------|-------------------|--------------------|--------------------|---------------------|-------------------|
| Primary amines (HC)        | 3.8 <sup>8</sup>  | 7.1 <sup>8</sup>  | 14.7 <sup>6</sup>  | 5.7 <sup>7</sup>  | 7.3 <sup>8</sup>  | 21.9 <sup>8</sup>  | 6.8 <sup>7</sup>  | 7.6 <sup>8</sup>   | 8.6 <sup>8</sup>   | 8.0 <sup>8</sup>    | 2.8 <sup>8</sup>  |
| Secondary amines (HC)      | 3.5 <sup>9</sup>  | 4.3 <sup>9</sup>  | 14.3 <sup>9</sup>  | 3.1 <sup>9</sup>  | 4.5 <sup>9</sup>  | 19.9 <sup>7</sup>  | 2.9 <sup>9</sup>  | 19.6 <sup>9</sup>  | 5.9 <sup>9</sup>   | 5.2 <sup>7</sup>    | 5.2 <sup>9</sup>  |
| Tertiary amines (HC)       | 5.5 <sup>4</sup>  | 3.5 <sup>4</sup>  | 13.9 <sup>4</sup>  | 2.8 <sup>4</sup>  | 3.5 <sup>4</sup>  | 16.1 <sup>4</sup>  | 4.7 <sup>4</sup>  | 23.2 <sup>4</sup>  | 3.6 <sup>4</sup>   | 3.6 <sup>4</sup>    | 8.3 <sup>4</sup>  |
| Amines                     | 3.2 <sup>24</sup> | 9.8 <sup>24</sup> | 19.6 <sup>24</sup> | 7.7 <sup>22</sup> | 9.1 <sup>24</sup> | 28.8 <sup>23</sup> | 7.6 <sup>23</sup> | 17.3 <sup>23</sup> | 11.5 <sup>24</sup> | 11.0 <sup>20</sup>  | 5.4 <sup>24</sup> |
| N in 6-membered rings (HC) | 5.0 <sup>12</sup> | 6.3 <sup>12</sup> | 38.1 <sup>11</sup> | 1.0 <sup>10</sup> | 3.7 <sup>11</sup> | 26.8 <sup>12</sup> | 5.7 <sup>12</sup> | 6.6 <sup>1</sup>   |                    | 16.1 <sup>127</sup> | 8.9 <sup>12</sup> |
| Cyanides (HC)              | 5.9 <sup>7</sup>  | 7.4 <sup>7</sup>  | 6.2 <sup>5</sup>   | 10.5 <sup>7</sup> | 8.2 <sup>7</sup>  | 58.8 <sup>7</sup>  | 9.7 <sup>6</sup>  |                    | 8.5 <sup>7</sup>   | 10.7                | 7.9 <sup>7</sup>  |
| Di-amides (HC)             | 2.0 <sup>2</sup>  |                   |                    |                   | 9.9 <sup>1</sup>  |                    |                   |                    | 2.6 <sup>1</sup>   |                     | 11.7 <sup>1</sup> |
| Nitrates (HC)              | 0.0 <sup>1</sup>  | 0.3 <sup>1</sup>  |                    | 0.4 <sup>1</sup>  | 0.3 <sup>1</sup>  |                    | 0.5 <sup>1</sup>  |                    | 0.3 <sup>1</sup>   | 0.6 <sup>1</sup>    | 0.6 <sup>1</sup>  |
| Nitrogenated (w/o amines)  | 6.0 <sup>33</sup> | 6.5 <sup>22</sup> | 28.1 <sup>16</sup> | 5.0 <sup>18</sup> | 5.7 <sup>19</sup> | 35.7 <sup>22</sup> | 6.7 <sup>21</sup> | 6.6 <sup>1</sup>   | 7.8 <sup>8</sup>   | 16.7 <sup>26</sup>  | 8.3 <sup>23</sup> |

Table 33

Critical pressure average absolute error (kPa) of the models for different types of nitrogen compounds (number of components given as superscript)

|                            | This work         | JR                | CG                | MP                | AB                | WQ                | DB                | LD                | SJ                | KR                |
|----------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Primary amines (HC)        | 278 <sup>8</sup>  | 437 <sup>8</sup>  | 282 <sup>6</sup>  | 193 <sup>7</sup>  | 402 <sup>8</sup>  | 383 <sup>7</sup>  | 464 <sup>8</sup>  | 380 <sup>8</sup>  | 342 <sup>8</sup>  | 544 <sup>8</sup>  |
| Secondary amines (HC)      | 128 <sup>8</sup>  | 338 <sup>8</sup>  | 398 <sup>8</sup>  | 281 <sup>8</sup>  | 305 <sup>8</sup>  | 315 <sup>8</sup>  | 445 <sup>8</sup>  | 356 <sup>8</sup>  | 366 <sup>7</sup>  | 257 <sup>8</sup>  |
| Tertiary amines (HC)       | 101 <sup>4</sup>  | 111 <sup>4</sup>  | 177 <sup>4</sup>  | 52 <sup>4</sup>   | 132 <sup>4</sup>  | 198 <sup>4</sup>  | 509 <sup>4</sup>  | 118 <sup>4</sup>  | 138 <sup>4</sup>  | 185 <sup>4</sup>  |
| Amines                     | 148 <sup>25</sup> | 467 <sup>25</sup> | 483 <sup>23</sup> | 400 <sup>21</sup> | 508 <sup>23</sup> | 556 <sup>22</sup> | 493 <sup>22</sup> | 472 <sup>25</sup> | 496 <sup>22</sup> | 485 <sup>25</sup> |
| N in 6-membered rings (HC) | 119 <sup>3</sup>  | 85 <sup>3</sup>   | 165 <sup>2</sup>  | 113 <sup>3</sup>  | 40 <sup>3</sup>   | 121 <sup>3</sup>  |                   |                   | 241 <sup>3</sup>  | 128 <sup>3</sup>  |
| Cyanides (HC)              | 157 <sup>6</sup>  | 176 <sup>6</sup>  | 226 <sup>5</sup>  | 152 <sup>6</sup>  | 269 <sup>6</sup>  | 149 <sup>5</sup>  |                   | 253 <sup>6</sup>  | 250 <sup>6</sup>  | 218 <sup>6</sup>  |
| Di-amides (HC)             | 0 <sup>1</sup>    |                   |                   |                   |                   |                   |                   |                   |                   |                   |
| Nitrates (HC)              | 0 <sup>1</sup>    | 410 <sup>1</sup>  |                   | 20 <sup>1</sup>   | 18 <sup>1</sup>   | 1 <sup>1</sup>    |                   | 40 <sup>1</sup>   | 407 <sup>1</sup>  | 437 <sup>1</sup>  |
| Nitrogenated (w/o amines)  | 100 <sup>13</sup> | 263 <sup>12</sup> | 208 <sup>7</sup>  | 177 <sup>10</sup> | 192 <sup>9</sup>  | 313 <sup>11</sup> |                   | 253 <sup>6</sup>  | 255 <sup>12</sup> | 176 <sup>12</sup> |

Table 34

Critical volume average relative error (%) of the models for different types of nitrogen compounds (number of components given as superscript)

|                            | This work         | JR                | CG                | MP               | AB                | WQ                | LD                | SJ                | KR                |
|----------------------------|-------------------|-------------------|-------------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Primary amines (HC)        | 2.6 <sup>5</sup>  | 6.8 <sup>5</sup>  | 4.2 <sup>3</sup>  | 3.3 <sup>3</sup> | 6.7 <sup>5</sup>  | 4.1 <sup>4</sup>  | 7.4 <sup>5</sup>  | 6.0 <sup>5</sup>  | 4.8 <sup>5</sup>  |
| Secondary amines (HC)      | 0.8 <sup>3</sup>  | 5.8 <sup>3</sup>  | 4.2 <sup>3</sup>  | 3.9 <sup>3</sup> | 9.5 <sup>3</sup>  | 4.4 <sup>3</sup>  | 5.0 <sup>3</sup>  | 7.7 <sup>2</sup>  | 7.5 <sup>3</sup>  |
| Tertiary amines (HC)       | 1.7 <sup>1</sup>  | 1.1 <sup>1</sup>  | 1.4 <sup>1</sup>  | 1.2 <sup>1</sup> | 1.8 <sup>1</sup>  | 1.2 <sup>1</sup>  | 4.6 <sup>1</sup>  | 2.1 <sup>1</sup>  | 2.8 <sup>1</sup>  |
| Amines                     | 1.4 <sup>12</sup> | 4.6 <sup>12</sup> | 9.1 <sup>10</sup> | 3.2 <sup>8</sup> | 7.7 <sup>12</sup> | 5.8 <sup>11</sup> | 6.1 <sup>12</sup> | 5.4 <sup>9</sup>  | 7.5 <sup>12</sup> |
| N in 6-membered rings (HC) | 3.1 <sup>4</sup>  | 4.3 <sup>4</sup>  | 5.4 <sup>3</sup>  | 6.1 <sup>3</sup> | 3.9 <sup>4</sup>  | 8.7 <sup>4</sup>  |                   | 4.8 <sup>4</sup>  | 6.1 <sup>4</sup>  |
| Cyanides (HC)              | 0.0 <sup>1</sup>  | 0.0 <sup>1</sup>  | 0.2 <sup>1</sup>  | 1.0 <sup>1</sup> | 0.3 <sup>1</sup>  | 0.9 <sup>1</sup>  | 0.3 <sup>1</sup>  | 0.2 <sup>1</sup>  | 2.5 <sup>1</sup>  |
| Di-amides (HC)             | 0.0 <sup>1</sup>  |                   |                   |                  | 10.2 <sup>1</sup> |                   | 1.1 <sup>1</sup>  |                   | 2.7 <sup>1</sup>  |
| Nitrates (HC)              | 0.0 <sup>1</sup>  | 0.1 <sup>1</sup>  |                   | 0.1 <sup>1</sup> | 0.2 <sup>1</sup>  | 0.1 <sup>1</sup>  | 0.2 <sup>1</sup>  | 0.2 <sup>1</sup>  | 0.2 <sup>1</sup>  |
| Nitrogenated (w/o amines)  | 2.6 <sup>18</sup> | 3.3 <sup>8</sup>  | 4.1 <sup>4</sup>  | 5.7 <sup>5</sup> | 4.4 <sup>6</sup>  | 10.7 <sup>8</sup> | 0.7 <sup>2</sup>  | 6.3 <sup>17</sup> | 5.0 <sup>9</sup>  |

Table 35

Critical temperature average absolute error (K) of the models for various other compounds (number of components given as superscript)

|                              | This work         | JR | CG | MP | AB | CT | WQ                 | DB | LD                 | SJ                | KR |
|------------------------------|-------------------|----|----|----|----|----|--------------------|----|--------------------|-------------------|----|
| Tin (HC)                     | 0.0 <sup>1</sup>  |    |    |    |    |    | 44.6 <sup>1</sup>  |    |                    | 0.9 <sup>1</sup>  |    |
| Boron (HC)                   | 0.0 <sup>1</sup>  |    |    |    |    |    | 119.0 <sup>1</sup> |    | 11.3 <sup>1</sup>  | 6.5 <sup>1</sup>  |    |
| Silicon                      | 2.6 <sup>2</sup>  |    |    |    |    |    |                    |    | 10.7 <sup>2</sup>  | 8.9 <sup>2</sup>  |    |
| Silicon to oxygen            | 9.5 <sup>16</sup> |    |    |    |    |    |                    |    | 16.0 <sup>13</sup> | 9.5 <sup>11</sup> |    |
| Silicon to either O, F or Cl | 5.9 <sup>7</sup>  |    |    |    |    |    |                    |    | 4.4 <sup>6</sup>   | 5.7 <sup>6</sup>  |    |

Table 36

Critical pressure average absolute error (kPa) of the models for various other types of compounds (number of components given as superscript)

|                              | This work        | JR | CG | MP | AB | WQ                | DB | LD                | SJ               | KR |
|------------------------------|------------------|----|----|----|----|-------------------|----|-------------------|------------------|----|
| Tin (HC)                     | 0 <sup>1</sup>   |    |    |    |    |                   |    |                   | 520 <sup>1</sup> |    |
| Boron (HC)                   | 0 <sup>1</sup>   |    |    |    |    | 2470 <sup>1</sup> |    | 1238 <sup>1</sup> | 703 <sup>1</sup> |    |
| Silicon                      | 147 <sup>2</sup> |    |    |    |    |                   |    | 108 <sup>2</sup>  | 132 <sup>2</sup> |    |
| Silicon to oxygen            | 53 <sup>13</sup> |    |    |    |    |                   |    | 74 <sup>13</sup>  | 79 <sup>9</sup>  |    |
| Silicon to either O, F or Cl | 58 <sup>10</sup> |    |    |    |    |                   |    | 114 <sup>9</sup>  | 147 <sup>9</sup> |    |

Table 37

Critical volume average relative error (%) of the models for various other types of compounds (number of components given as superscript)

|                              | This work        | JR | CG | MP | AB | WQ | LD                | SJ               | KR |
|------------------------------|------------------|----|----|----|----|----|-------------------|------------------|----|
| Silicon                      | 0.0 <sup>1</sup> |    |    |    |    |    | 28.0 <sup>1</sup> | 4.0 <sup>1</sup> |    |
| Silicon to oxygen            | 2.0 <sup>8</sup> |    |    |    |    |    | 41.8 <sup>8</sup> | 4.4 <sup>8</sup> |    |
| Silicon to either O, F or Cl | 2.6 <sup>7</sup> |    |    |    |    |    | 24.9 <sup>6</sup> | 2.2 <sup>6</sup> |    |

non-additive behavior are summarized in Tables 38–40. In these cases, all other methods produce huge errors as they assume additivity.

#### 4. Database

The quality of a method strongly depends on the amount and quality of the experimental data used in its development. The critical property data set finally used in this work after extensive data evaluation (see Section 5) consisted of 588 critical temperatures, 486 critical pressures and 348 critical volumes from the Dortmund Data Bank (DDB [31]), all of which are constructed from the structural groups proposed in this work.

The required primary database (DDB-pure) was built up since 1991 at the University of Oldenburg in Germany in cooperation

with groups in Prague, Tallinn, Minsk, Berlin and Graz and currently (2006) contains properties for 16 000 components from 19 700 references (1 048 000 data tuples<sup>2</sup>).

Critical temperature data from the DDB were only used for the training set if experimental normal boiling temperatures were also available.

The frequency of occurrence of the individual structural groups and second-order corrections in the database for the model development is given in Tables 7–15.

<sup>2</sup> In mathematics, a tuple is a finite sequence (also known as an “ordered list”) of objects, each of a specified type. A tuple containing  $n$  objects is known as an “ $n$ -tuple”. For example, the 4-tuple (or “quadruple”), with components of respective types Person, Day, Month and Year, could be used to record that a certain person was born on a certain day of a certain month of a certain year (<http://www.wikipedia.org>).

Table 38

Critical temperature average absolute error (K) of the models for multifunctional compounds (number of components given as superscript)

|                    | This work         | JR                 | CG                 | MP                 | AB                 | CT                 | WQ                 | DB                 | LD                 | SJ                 | KR                 |
|--------------------|-------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| OH                 | 5.8 <sup>17</sup> | 42.6 <sup>17</sup> | 50.9 <sup>17</sup> | 37.8 <sup>17</sup> | 21.9 <sup>17</sup> | 51.7 <sup>17</sup> | 36.6 <sup>17</sup> | 95.3 <sup>17</sup> | 44.9 <sup>17</sup> | 39.5 <sup>13</sup> | 23.3 <sup>17</sup> |
| OH(a)              | 0.0 <sup>1</sup>  | 4.7 <sup>1</sup>   | 35.8 <sup>1</sup>  | 15.4 <sup>1</sup>  | 4.6 <sup>1</sup>   |                    | 28.1 <sup>1</sup>  | 40.5 <sup>1</sup>  | 53.0 <sup>1</sup>  | 2.0 <sup>1</sup>   | 14.4 <sup>1</sup>  |
| NH <sub>2</sub>    | 0.0 <sup>3</sup>  | 24.1 <sup>3</sup>  | 39.8 <sup>3</sup>  | 22.9 <sup>3</sup>  | 20.5 <sup>3</sup>  | 68.4 <sup>3</sup>  | 27.9 <sup>3</sup>  | 41.5 <sup>3</sup>  | 24.8 <sup>3</sup>  | 29.7 <sup>3</sup>  | 18.3 <sup>3</sup>  |
| NH                 | 0.0 <sup>2</sup>  | 14.8 <sup>2</sup>  | 27.3 <sup>2</sup>  | 15.8 <sup>2</sup>  | 7.0 <sup>2</sup>   | 53.0 <sup>1</sup>  | 6.5 <sup>2</sup>   | 35.6 <sup>2</sup>  | 21.7 <sup>2</sup>  | 25.7 <sup>1</sup>  | 4.9 <sup>2</sup>   |
| COOH               | 12.8 <sup>3</sup> |                    |                    |                    |                    |                    |                    |                    |                    | 0.4 <sup>1</sup>   |                    |
| Ether              | 7.4 <sup>44</sup> | 17.3 <sup>31</sup> | 20.8 <sup>27</sup> | 16.8 <sup>28</sup> | 9.8 <sup>30</sup>  | 30.7 <sup>29</sup> | 18.8 <sup>30</sup> | 92.4 <sup>25</sup> | 18.3 <sup>41</sup> | 15.6 <sup>32</sup> | 16.1 <sup>31</sup> |
| Epoxide            | 0.0 <sup>1</sup>  | 31.2 <sup>1</sup>  | 46.8 <sup>1</sup>  | 17.9 <sup>1</sup>  | 28.8 <sup>1</sup>  |                    | 29.9 <sup>1</sup>  | 24.3 <sup>1</sup>  | 26.7 <sup>1</sup>  |                    | 49.6 <sup>1</sup>  |
| Ester              | 1.5 <sup>8</sup>  | 3.9 <sup>8</sup>   | 8.8 <sup>8</sup>   | 3.5 <sup>7</sup>   | 3.5 <sup>8</sup>   | 13.7 <sup>8</sup>  | 7.5 <sup>8</sup>   | 87.4 <sup>7</sup>  | 4.7 <sup>8</sup>   | 3.3 <sup>7</sup>   | 14.9 <sup>8</sup>  |
| Ketone             | 0.0 <sup>1</sup>  |                    |                    |                    |                    | 157.0 <sup>1</sup> | 46.9 <sup>1</sup>  |                    |                    |                    |                    |
| Teth, AtS, AO, AN5 | 6.9 <sup>6</sup>  | 5.0 <sup>3</sup>   |                    | 6.4 <sup>1</sup>   |                    | 17.1 <sup>3</sup>  | 41.1 <sup>3</sup>  |                    |                    | 22.9 <sup>3</sup>  | 5.8 <sup>3</sup>   |
| All GI components  | 6.7 <sup>64</sup> | 21.2 <sup>47</sup> | 33.4 <sup>39</sup> | 21.4 <sup>41</sup> | 14.6 <sup>42</sup> | 43.6 <sup>44</sup> | 21.5 <sup>46</sup> | 84.6 <sup>36</sup> | 23.5 <sup>54</sup> | 18.2 <sup>46</sup> | 18.4 <sup>47</sup> |

Table 39

Critical pressure average absolute error (K) deviations of the models for different types of multifunctional compounds (number of components given as superscript)

|                    | This work         | JR                | CG                | MP                | AB                | WQ                | DB                | LD                | SJ                | KR                |
|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| OH                 | 128 <sup>17</sup> | 608 <sup>17</sup> | 972 <sup>17</sup> | 566 <sup>17</sup> | 851 <sup>17</sup> | 500 <sup>17</sup> | 597 <sup>17</sup> | 589 <sup>17</sup> | 735 <sup>13</sup> | 592 <sup>17</sup> |
| OH(a)              | 0 <sup>1</sup>    | 572 <sup>1</sup>  | 198 <sup>1</sup>  | 172 <sup>1</sup>  | 116 <sup>1</sup>  | 213 <sup>1</sup>  | 335 <sup>1</sup>  | 446 <sup>1</sup>  | 86 <sup>1</sup>   | 113 <sup>1</sup>  |
| NH <sub>2</sub>    | 0 <sup>3</sup>    | 1668 <sup>3</sup> | 1575 <sup>3</sup> | 1672 <sup>3</sup> | 1930 <sup>3</sup> | 1727 <sup>3</sup> | 1501 <sup>3</sup> | 1839 <sup>3</sup> | 1788 <sup>3</sup> | 1767 <sup>3</sup> |
| NH                 | 0 <sup>2</sup>    | 107 <sup>2</sup>  | 481 <sup>2</sup>  | 114 <sup>2</sup>  | 348 <sup>2</sup>  | 151 <sup>2</sup>  | 295 <sup>2</sup>  | 210 <sup>2</sup>  | 25 <sup>1</sup>   | 245 <sup>2</sup>  |
| Ether              | 73 <sup>33</sup>  | 353 <sup>23</sup> | 328 <sup>19</sup> | 250 <sup>20</sup> | 226 <sup>22</sup> | 319 <sup>22</sup> | 316 <sup>20</sup> | 197 <sup>33</sup> | 258 <sup>26</sup> | 318 <sup>23</sup> |
| Epoxide            | 0 <sup>1</sup>    | 737 <sup>1</sup>  | 296 <sup>1</sup>  | 126 <sup>1</sup>  | 245 <sup>1</sup>  | 139 <sup>1</sup>  | 180 <sup>1</sup>  | 351 <sup>1</sup>  |                   | 920 <sup>1</sup>  |
| Ester              | 74 <sup>7</sup>   | 150 <sup>7</sup>  | 136 <sup>7</sup>  | 204 <sup>6</sup>  | 134 <sup>7</sup>  | 188 <sup>7</sup>  | 538 <sup>6</sup>  | 135 <sup>7</sup>  | 155 <sup>6</sup>  | 155 <sup>7</sup>  |
| Ketone             | 0 <sup>1</sup>    | 306 <sup>1</sup>  |                   | 121 <sup>1</sup>  |                   | 130 <sup>1</sup>  |                   | 279 <sup>1</sup>  | 509 <sup>1</sup>  | 50 <sup>1</sup>   |
| Teth, AtS, AO, AN5 | 0 <sup>5</sup>    | 600 <sup>5</sup>  |                   | 514 <sup>1</sup>  |                   | 1030 <sup>5</sup> |                   | 181 <sup>2</sup>  | 285 <sup>5</sup>  | 207 <sup>5</sup>  |
| All GI components  | 77 <sup>50</sup>  | 488 <sup>40</sup> | 650 <sup>31</sup> | 401 <sup>34</sup> | 530 <sup>34</sup> | 451 <sup>39</sup> | 536 <sup>31</sup> | 323 <sup>47</sup> | 378 <sup>40</sup> | 406 <sup>40</sup> |

## 5. Test of the predictive capability

To test the predictive capability of a correlation, it is common to estimate values for a test set of data not used in the regression. This procedure allows one to verify both the applicability of the model and the reliability of the data used to regress the model parameters. In the case of critical property data, only a rather small set of experimental data is available. This was the reason why only one limited test of this type was conducted in this work. It should also be noted that the Ambrose method did not use this form of test. Instead, the author proceeded very carefully to ensure that the different model parameters were set to physically realistic values.

To ensure their validity, experimental critical property data were carefully examined and partly revised within the devel-

opment of a generally applicable volume translated group contribution equation of state and within the development of pure component data correlations for process control. During this work available data were verified using a large amount of other types of data like vapor pressure, density, heat of vaporization, heat capacity, etc. Discussions concerning the reliability of experimental critical data in well-known review articles (e.g., [32]) supplied an additional point of view.

Group definitions were taken from the previously published normal boiling temperature method with minor improvements. Thereby it can be assumed that the method contains only the necessary differentiations for the estimation of critical property data. Unnecessary or redundant differentiations introduce partly linear dependent parameters and usually lead to unrealistic parameter values.

Table 40

Critical volume average relative error (%) of the models for different types of multifunctional compounds (number of components given as superscript)

|                    | This work         | JR                | CG                 | MP                 | AB                | WQ                 | LD                 | SJ                | KR                |
|--------------------|-------------------|-------------------|--------------------|--------------------|-------------------|--------------------|--------------------|-------------------|-------------------|
| OH                 | 0.7 <sup>5</sup>  | 4.6 <sup>5</sup>  | 6.7 <sup>5</sup>   | 9.0 <sup>4</sup>   | 5.2 <sup>5</sup>  | 7.4 <sup>5</sup>   | 4.6 <sup>5</sup>   | 4.9 <sup>5</sup>  | 10.1 <sup>5</sup> |
| OH(a)              | 0.0 <sup>1</sup>  | 27.7 <sup>1</sup> | 16.5 <sup>1</sup>  |                    | 12.3 <sup>1</sup> | 7.7 <sup>1</sup>   | 11.3 <sup>1</sup>  | 15.7 <sup>1</sup> | 8.7 <sup>1</sup>  |
| NH                 | 0.0 <sup>2</sup>  | 0.5 <sup>2</sup>  | 13.7 <sup>2</sup>  | 2.9 <sup>1</sup>   | 10.2 <sup>2</sup> | 6.6 <sup>2</sup>   | 1.8 <sup>2</sup>   | 1.0 <sup>1</sup>  | 14.4 <sup>2</sup> |
| OCN                | 3.9 <sup>4</sup>  |                   |                    |                    |                   |                    |                    | 5.5 <sup>4</sup>  |                   |
| Ether              | 2.5 <sup>19</sup> | 7.4 <sup>12</sup> | 9.6 <sup>9</sup>   | 7.7 <sup>10</sup>  | 8.4 <sup>10</sup> | 8.6 <sup>12</sup>  | 20.3 <sup>19</sup> | 5.3 <sup>17</sup> | 7.8 <sup>12</sup> |
| Epoxide            | 0.0 <sup>1</sup>  | 31.9 <sup>1</sup> | 38.7 <sup>1</sup>  |                    | 35.4 <sup>1</sup> | 16.4 <sup>1</sup>  | 34.1 <sup>1</sup>  |                   | 35.9 <sup>1</sup> |
| Ester              | 3.1 <sup>3</sup>  | 3.9 <sup>3</sup>  | 6.5 <sup>3</sup>   | 5.0 <sup>3</sup>   | 4.3 <sup>3</sup>  | 4.4 <sup>3</sup>   | 4.4 <sup>3</sup>   | 4.6 <sup>3</sup>  | 2.6 <sup>3</sup>  |
| Ketone             | 0.0 <sup>1</sup>  | 43.0 <sup>1</sup> |                    | 57.0 <sup>1</sup>  |                   | 46.8 <sup>1</sup>  | 40.0 <sup>1</sup>  | 37.8 <sup>1</sup> | 48.2 <sup>1</sup> |
| Teth, AtS, AO, AN5 | 0.0 <sup>5</sup>  | 4.4 <sup>5</sup>  |                    | 9.4 <sup>1</sup>   |                   | 19.6 <sup>5</sup>  | 3.7 <sup>2</sup>   | 4.0 <sup>5</sup>  | 3.3 <sup>5</sup>  |
| All GI components  | 2.0 <sup>32</sup> | 8.6 <sup>21</sup> | 11.0 <sup>13</sup> | 10.9 <sup>14</sup> | 8.7 <sup>14</sup> | 12.0 <sup>21</sup> | 18.0 <sup>25</sup> | 6.5 <sup>29</sup> | 9.3 <sup>21</sup> |

Table 41  
Critical property deviations for all models

| Methods                     | $T_c$                |                                   |                            | $P_c$                |                                   |                              | $V_c$                |                                   |                                                     |
|-----------------------------|----------------------|-----------------------------------|----------------------------|----------------------|-----------------------------------|------------------------------|----------------------|-----------------------------------|-----------------------------------------------------|
|                             | Number of components | Average absolute percentage error | Absolute average error (K) | Number of components | Average absolute percentage error | Absolute average error (kPa) | Number of components | Average absolute percentage error | Absolute average error ( $\text{cm}^3/\text{mol}$ ) |
| This work                   | 588                  | 0.74                              | 4.31                       | 486                  | 2.96                              | 99.82                        | 348                  | 1.79                              | 6.40                                                |
| Ambrose                     | 528                  | 1.07                              | 6.00                       | 412                  | 7.03                              | 252.46                       | 242                  | 3.99                              | 19.41                                               |
| Marrero Pardillo            | 458                  | 1.21                              | 7.79                       | 381                  | 6.04                              | 208.92                       | 248                  | 3.36                              | 16.11                                               |
| Klinevicz/Reid              | 547                  | 1.27                              | 7.83                       | 452                  | 7.57                              | 245.67                       | 319                  | 3.96                              | 17.87                                               |
| Wen/Qiang (Tb)              | 506                  | 1.26                              | 7.79                       |                      |                                   |                              |                      |                                   |                                                     |
| Joback                      | 543                  | 1.41                              | 8.75                       | 452                  | 7.11                              | 238.20                       | 314                  | 3.73                              | 16.49                                               |
| Somayajulu                  | 517                  | 1.44                              | 8.39                       | 438                  | 9.51                              | 294.59                       | 307                  | 4.14                              | 20.08                                               |
| Lydersen                    | 557                  | 1.71                              | 10.65                      | 474                  | 7.07                              | 227.53                       | 327                  | 5.37                              | 30.67                                               |
| Daubert                     | 475                  | 3.87                              | 23.91                      | 352                  | 7.00                              | 252.45                       |                      |                                   |                                                     |
| Methods not requiring $T_b$ |                      |                                   |                            |                      |                                   |                              |                      |                                   |                                                     |
| Wen/Qiang                   | 506                  | 2.97                              | 16.73                      | 421                  | 5.67                              | 196.81                       | 294                  | 4.99                              | 22.05                                               |
| Gani, Constantinou          | 559                  | 4.07                              | 17.19                      | 410                  | 7.12                              | 247.84                       | 277                  | 4.81                              | 22.84                                               |
| Chein-Hsiun Tu              | 572                  | 4.26                              | 23.31                      |                      |                                   |                              |                      |                                   |                                                     |

In the case of critical temperatures, data for 67 components were not used in the training set due to the lack of reliable experimental information on the normal boiling temperature. When estimating the critical temperatures for these components using estimated  $T_b$ -values from our previous method, an average absolute deviation of 10.87 K (1.93%) was obtained. This compares favourably to other methods which do not require the normal boiling temperature (CG, 19.9 K for 56 components; CT, 39.8 K for 58 components; and WQ1 (similar to WQ, here the model does not require  $T_b$ ), 14.5 K for 52 components).

The absolute average deviation of the boiling temperature method was 6.52 K for all 2812 components in the training set and 6.37 K for 199 components in the test set [21].

Estimating all critical temperature data in the training set using estimated instead of experimental  $T_b$ -values increases the absolute average deviation by about 5 K from 4.31 K (0.74%) to 9.21 K (1.65%). Based on these findings, the deviation in critical temperature for the test set is probably around 6 K while another 5 K can be attributed to the error in normal boiling temperature estimation. This compares well to the deviation of 4.31 K in the training set as the test set contains numerous data for exotic compounds and of questionable accuracy.

## 6. Discussion

### 6.1. Performance of the new model compared to other models, recommendations and limitations

The average absolute percentage error of the new method is in all cases significantly lower than that of the other methods (see Table 41). The new method combines the lowest deviation with the broadest range of applicability (i.e., number of components for which the property can be estimated).

The method, apart from being more accurate than previous methods, also has a significantly improved probability of predicting properties within a smaller prediction error as is shown in Figs. 2–4. From our point of view, these graphs serve as the most important criterion for the reliability of a model. The curves in Figs. 2–4 were calculated using all estimations possible for each method and were not limited to a common set of data.

In the case of critical temperature, only 10% of the data are estimated with a deviation greater than 10 K and approximately, 2.5% with a deviation greater than 20 K. The large deviations for CT and CG in Fig. 2 are partly caused by the fact that these methods estimate the critical temperature without information about the normal boiling temperature.

Besides the average deviations of the models and the information from Figs. 2–4, one should also be aware of the problems of JR, LD, MP, DB and KR with respect to extrapolation to large molecules (Fig. 1).

The authors can only recommend, besides the new method presented in this paper, the methods of Ambrose (AB) and to a lesser extent KR (for not too large molecules) for critical property data estimation. The other methods should be used for multi-functional compounds with great caution.

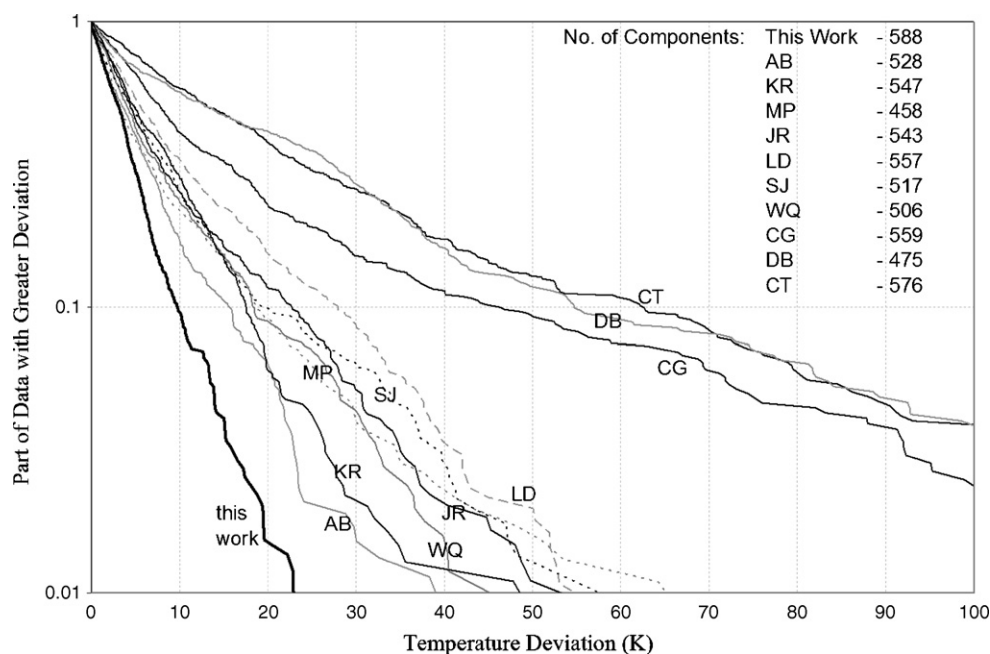


Fig. 2. Fraction of the data with deviations for critical temperature larger than a given temperature.

Due to the limited number of experimental data, not all of the required group increments could be regressed. This limits the applicability of the method to component classes for which data are available but at the same time improves the reliability. Most of the missing parameters are group interaction parameters. Discarding group interaction contributions for polyols increases the average absolute deviation in critical temperature from 5.8 to 21.6 K, which is similar to the deviations of AB and KR (see Table 38). All other models perform significantly worse. For this reason the new model can also be recommended in cases where

group interaction parameters are not available. In this case one should expect a higher uncertainty in the result.

Several model parameters are based on very limited experimental information. In Tables 7–15, some of the parameters are enclosed in brackets. For these cases the parameter values created serious doubt about their validity. In the case of the OCN–OCN group interaction contribution (number 176) for the critical temperature estimation all three data points were measured by the same author [27] and the parameter value is physically very improbable.

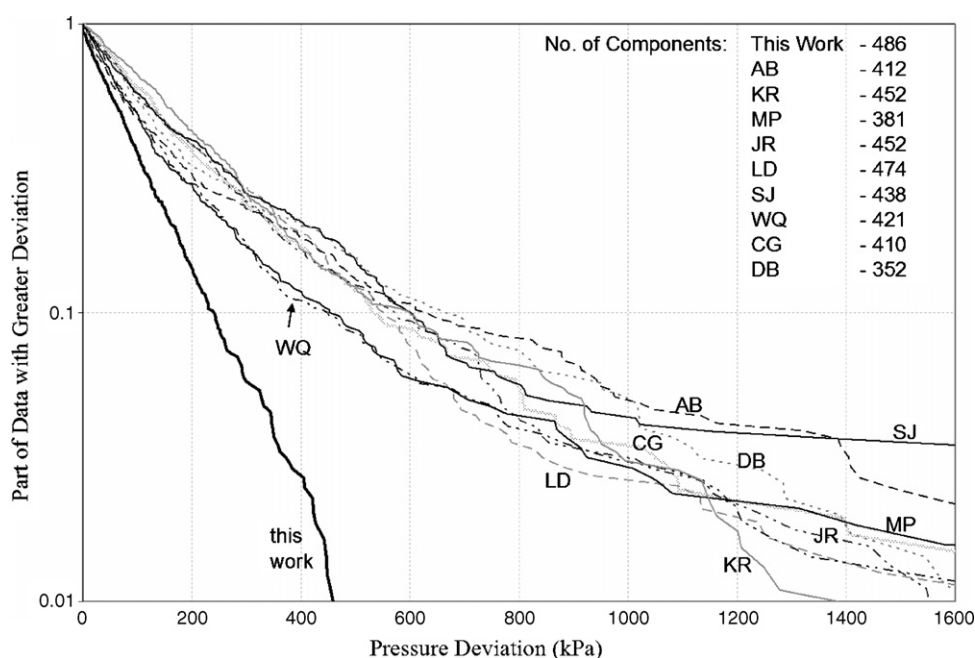


Fig. 3. Fraction of the data with deviations for critical pressure larger than a given pressure.

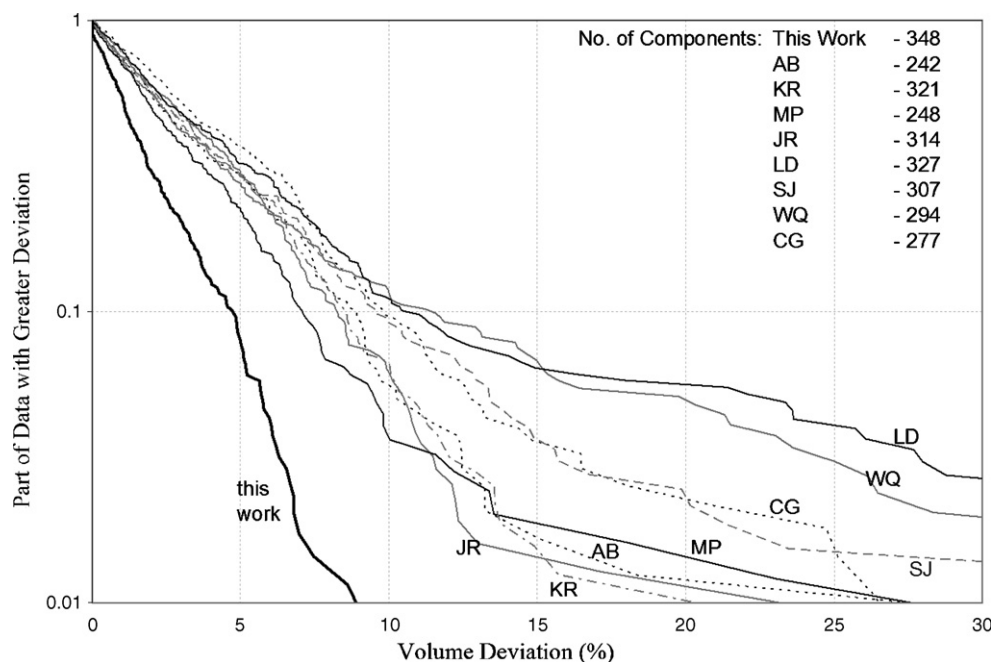


Fig. 4. Fraction of the data with deviations for critical volume larger than a given percentage volume deviation.

## 6.2. Cases where larger deviations can be expected

Large deviations were observed for small components, which are among the first in a homologous series. It is often the case that first members in a homologues series do not follow the trend of the series. Fortunately, experimental data are mostly available for these simple substances and thus, there is no need for group contribution estimation. Typical examples of larger deviations (experimental minus calculated value) are:

|                     |                        |                         |                        |
|---------------------|------------------------|-------------------------|------------------------|
| Ethane              | $\Delta T_c = -9.8$ K  | $\Delta P_c = -210$ kPa | $\Delta V_c = -5.2\%$  |
| Propane             | $\Delta T_c = -9.2$ K  | $\Delta P_c = -152$ kPa | $\Delta V_c = +1.2\%$  |
| Glycerol            | $\Delta T_c = +30.9$ K | $\Delta P_c = -170$ kPa |                        |
| 2,2-Diethoxypropane | $\Delta T_c = -30.0$ K |                         |                        |
| Allyl sulfide       | $\Delta T_c = +23.8$ K |                         |                        |
| Dimethoxymethane    | $\Delta T_c = +16.2$ K | $\Delta P_c = -262$ kPa | $\Delta V_c = -11.5\%$ |
| Dimethylamine       | $\Delta T_c = -12.7$ K | $\Delta P_c = -20$ kPa  |                        |

In some cases large deviations could be explained by reviewing the original reference. Estimation of the critical pressure of 1,3-propanediol resulted in an error of 422 kPa (the largest error observed for multi-functional alkanols). The author stated an experimental absolute error of 600 kPa, so even the large estimation error is still within the estimated accuracy of the experimental value. Whenever more reliable data were available, sources with high experimental uncertainty were removed from the training set.

## 7. Examples

To illustrate the application of the proposed method, a detailed procedure for the estimation of critical properties is given in [Appendix A \(Tables 42a–42d\)](#) for

- the critical temperature of 2,2,3,3-tetramethylbutane, illustrating the application of the steric correction;
- the critical pressure of diethylene glycol monomethyl ether, illustrating the application of group interactions;
- the critical volume of trichloro silane, illustrating the use of the new silicon groups and correction;
- Estimation of critical temperature of perfluoro-2-propanone, illustrating the application of the C=C–C=O correction.

## 8. Conclusion

A new group contribution method has been developed for the prediction of critical properties. The method performs significantly better than other methods used for comparison. In addition, it gives physically realistic extrapolations to larger molecules and has by far the largest range of applicability. The method is based on a previous method for the normal boiling point estimation with a few minor modifications. Structural groups were defined in a standardized form and the fragmentation of the molecular structures was performed by an automatic procedure to eliminate any arbitrary assumptions.

The performance of the new group contribution method compared to a large number of literature methods is reported for an extensive set of substance classes in order to give the user information about the expected accuracy. These tables also contain information on the number of components in the different classes. This information should be of value for experimentalists searching for substances that would add new knowledge about critical properties.

Most recent updates and corrections can be found at <http://www.property-estimation.org>.

## Acknowledgement

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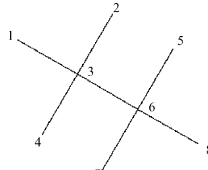
DDBST GmbH for providing data and software support for this project and J. Gmehling as project leader.

## Appendix A

See Tables 42a–42d for sample calculations for estimation of critical temperature, pressure and volume.

Table 42a

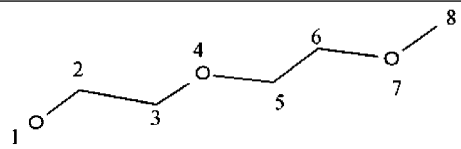
Estimation of the critical temperature of 2,2,3,3-tetramethylbutane

|  |                     |           |              |            |
|------------------------------------------------------------------------------------|---------------------|-----------|--------------|------------|
| Component: 2,2,3,3-tetramethylbutane; number of atoms: 8                           |                     |           |              |            |
| Group                                                                              | Atoms/bond          | Frequency | Contribution | Total      |
| 1                                                                                  | 1,2,4,5,7,8 (Atoms) | 6         | 0.0418682    | 0.2512092  |
| 6                                                                                  | 3,6 (atoms)         | 2         | −0.0038778   | −0.0077556 |
| Steric corrections                                                                 |                     |           |              |            |
| C3C–CC3 (133)                                                                      | 3–6 (bond)          | 1         | −0.0245802   | −0.0245802 |
| Total sum                                                                          |                     |           |              | 0.2188734  |

$$T_c = 379.6 \text{ K} \left( 0.6990 + \frac{1}{0.9889 + 0.2188734^{0.8607}} \right) = 566.8 \text{ K}, T_{c,\text{exp}} = 567.8 \text{ K}.$$

Table 42b

Estimation of the critical pressure of diethylene glycol monomethyl ether

|        |          |                     |              |              |
|--------------------------------------------------------------------------------------------|----------|---------------------|--------------|--------------|
| Component: diethylene glycol monomethyl ether; number of atoms:8; molecular weight: 120.15 |          |                     |              |              |
| Group                                                                                      | Atoms    | Frequency           | Contribution | Total        |
| 2                                                                                          | 8        | 1                   | 0.00055262   | 0.00055262   |
| 7                                                                                          | 2,3,5,6  | 4                   | 0.00034310   | 0.00137240   |
| 35                                                                                         | 1        | 1                   | −0.00018153  | −0.00018153  |
| 38                                                                                         | 4,7      | 2                   | 0.00020592   | 0.00041184   |
| Interactions                                                                               |          |                     |              |              |
| OH–Ether (140)                                                                             | 1–4, 1–7 | (2 <sup>a</sup> )/8 | 0.00073373   | 0.0001834325 |
| Ether–Ether (178)                                                                          | 4–7      | (1 <sup>a</sup> )/8 | 0.00026751   | 0.0000334388 |
| Total sum (rounded)                                                                        |          |                     |              | 0.00237220   |

$$P_c = \frac{120.15^{-0.14041}}{(0.00939 + 0.00237220)^2} = 3689.9 \text{ kPa}, P_{c,\text{exp}} = 3670.0 \text{ kPa}.$$

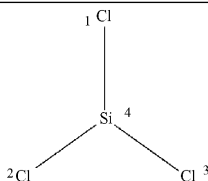
<sup>a</sup> From Eq. (4):  $m = 3$  (one OH and two EtherO);

$$GI = (COH(1)–EtherO(4) + COH(1)–EtherO(7) + CEtherO(4)–OH(1) + CEtherO(7)–OH(1) + CEtherO(4)–EtherO(7) + CEtherO(7)–EtherO(4))/(n \times (m - 1))$$

$$= (2CEtherO–EtherO + 4COH–EtherO)/(n \times (m - 1)) = (CEtherO–EtherO + 2COH–EtherO)/n.$$

Table 42c

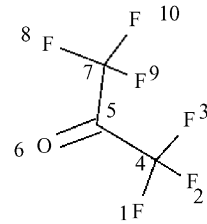
Estimation of the critical volume of trichloro silane

| Component: trichloro silane; Number of atoms: 4 |       |  |              |          |
|-------------------------------------------------|-------|------------------------------------------------------------------------------------|--------------|----------|
| Group                                           | Atoms | Frequency                                                                          | Contribution | Total    |
| 27                                              | 1,2,3 | 3                                                                                  | 26.2582      | 78.7746  |
| 215                                             | 4     | 1                                                                                  | 37.0423      | 37.0423  |
| Corrections                                     |       |                                                                                    |              |          |
| 124                                             | –     | 1                                                                                  | –6.1909      | –6.1909  |
| 217                                             | 4     | 1                                                                                  | 19.4348      | 19.4348  |
| Total sum                                       |       |                                                                                    |              | 129.0608 |

$V_c = \frac{129.0608}{4-0.2266} + 86.1539 = 262.9 \text{ cm}^3 \text{ mol}^{-1}$ ,  $V_{c,\text{exp}} = 268.0 \text{ cm}^3 \text{ mol}^{-1}$ .

Table 42d

Estimation of critical temperature of perfluoro-2-propanone

| Component: perfluoro-2-propanone; number of atoms: 10 |              |  |              |            |
|-------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|--------------|------------|
| Group                                                 | Atoms        | Frequency                                                                           | Contribution | Total      |
| 7                                                     | 4,7          | 2                                                                                   | 0.0528003    | 0.1056006  |
| 21                                                    | 1,2,3,8,9,10 | 6                                                                                   | 0.0181302    | 0.1087812  |
| 51                                                    | 5,6          | 1                                                                                   | 0.0561572    | 0.0561572  |
| Corrections                                           |              |                                                                                     |              |            |
| 120                                                   | 5            | 1                                                                                   | 0.0114437    | 0.0114437  |
| 121                                                   | 4,7          | 2                                                                                   | –0.0013023   | –0.0026046 |
| 123                                                   | –            | 1                                                                                   | –0.0013798   | –0.0013798 |
| Total sum                                             |              |                                                                                     |              | 0.2779983  |

$T_c = 245.9 \text{ K} \left( 0.6990 + \frac{1}{0.9889 + 0.2779983^{0.8607}} \right) = 358.0 \text{ K}$ ,  $T_{c,\text{exp}} = 357.2 \text{ K}$ .

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